**A PROJECT REPORT ON**

# "CREDO SCORE: A LOAN DEFAULT RISK PREDICTOR"

*Submitted in partial fulfilment of the requirements for the*

*award of the degree*

**OF**

**BACHELOR OF COMPUTER APPLICATIONS (BCA)**

**To**

**GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, DELHI**


**SUBMITTED BY:**

**GURSIMRAN KAUR**

**02490202024**


UNDER THE SUPERVISION OF

MS. NISHA CHOUDHARY

(ASSISTANT PROFESSOR)





**SRI GURU TEGH BAHADUR INSTITUTE OF MANAGEMENT AND INFORMATION TECHNOLOGY**

*(24, GT Karnal Rd, Gurdwara Nanak Piao Campus, Kirpal Bagh, Gujranwala Town, New Delhi, Delhi, 110033)*


**(2024-2027)**

# CERTIFICATE

This is to certify that the Summer Training Report entitled "Loan Default Risk Predictor: An Explainable Machine Learning Web Application for Credit Risk Assessment" is a bonafide work carried out by GURSIMRAN KAUR, Enrolment No. 02490202024, a student of Bachelor of Computer Applications (BCA), Semester V, Section 5A, at Sri Guru Tegh Bahadur Institute of Management and Information Technology, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications.

The work was carried out under my guidance during the six-week summer training, and to the best of my knowledge it is original and has not been submitted elsewhere for any degree or diploma.

---

Ms. Nisha Choudhary
Mentor / Guide

---

Dr. Raj Kumar
HOD (BCA)

Date: _____ Place: _____

# COMPANY'S CERTIFICATE

Certificate of Completion issued by GRAStech to GURSIMRAN KAUR for successfully completing the six-week online internship program in Machine Learning Using Python. Date of issue: 31 July 2026. Signed by Mohammad Qayam Ali, Founder and Program Head.



# CANDIDATE'S DECLARATION

I, GURSIMRAN KAUR, Enrolment No. 02490202024, a student of BCA, Semester V, Section 5A, Sri Guru Tegh Bahadur Institute of Management and Information Technology, hereby declare that the Summer Training Report entitled "Loan Default Risk Predictor: An Explainable Machine Learning Web Application for Credit Risk Assessment" is my own original work, carried out during the six-week online internship in Machine Learning Using Python at GRAStech, and that it has been written in my own words.

I further declare that this report has not been copied from any book, website or another student, that wherever the work of others has been used it has been acknowledged in the references, and that this report has not been submitted to any other institute or university for the award of any degree or diploma.

GURSIMRAN KAUR
Enrolment No.: 02490202024

Date: _____ Place: _____

# ACKNOWLEDGEMENT

I express my sincere gratitude to everyone who helped me in completing this Summer Training Report and the project Loan Default Risk Predictor.

I am thankful to GRAStech and Mohammad Qayam Ali, Founder and Program Head, for providing the opportunity of a six-week online internship in Machine Learning Using Python, which gave me the practical exposure needed to build this project.

I am grateful to my mentor, Ms. Nisha Choudhary, Assistant Professor, for guiding me, reviewing the report and the working project, and giving valuable suggestions at every stage.

I thank the management and faculty of Sri Guru Tegh Bahadur Institute of Management and Information Technology for providing a good learning environment.

GURSIMRAN KAUR

# ABSTRACT

Lenders that extend consumer credit must decide, quickly and consistently, whether an applicant is likely to repay a loan, in a way that is explainable to the applicant and defensible to a regulator. Manual underwriting varies between officers, traditional scorecards struggle to capture non-linear effects among income, loan amount and credit history, and the machine learning models that do capture such effects usually behave as a black box, giving a probability with no reason a loan officer can examine. A missed default and a wrongly declined applicant also carry different costs, and few tools let an institution weigh the two against each other.

Loan Default Risk Predictor is a web application, developed during the six-week summer training (an online internship in Machine Learning Using Python at GRAStech), that scores an applicant's probability of default and explains the score factor by factor. A user enters twelve fields covering the applicant, the loan and the existing credit position, or loads a sample file, and receives a calibrated probability of default, a risk band, a score and a lending recommendation, along with a waterfall of the contribution each factor made to the score. A what-if control lets the user move the approval cutoff and see how the approval rate, the bad rate and the expected cost change.

The model is a gradient boosting ensemble of 250 trees, trained on the UCI German Credit dataset of 1,000 applicants, benchmarked against logistic regression, random forest and XGBoost under five-fold cross-validation, tuned by randomized search and Platt-calibrated so its output reads as a true probability. Contributions are computed with a Saabas tree-path decomposition, rescaled to sum exactly to the predicted probability. The application runs Next.js and React on the front end and a dependency-free Python scoring function on the back end, deployed as a Vercel serverless function, and was verified with 13 notebook tests, 25 API tests and 18 interface tests, all passing.

The report covers the problem formulation, system analysis, system design including the ERD, DFD and use case diagrams, the model's evaluation, testing, implementation, limitations, future scope, and annexures of screen snapshots and source code. The live application is available at https://credoscore-gk.vercel.app.

Keywords: Credit Risk, Probability of Default, Gradient Boosting, Platt Scaling, Explainable AI, Next.js, Python, Vercel.

# LIST OF FIGURES

Figure 2.1 Project schedule of the six-week training ..... 6
Figure 3.1 Three-tier architecture of the Loan Default Risk Predictor ..... 12
Figure 3.2 Module hierarchy of the Loan Default Risk Predictor ..... 13
Figure 3.3 Entity Relationship Diagram of the Loan Default Risk Predictor ..... 15
Figure 3.4 Context Level DFD (Level 0) ..... 18
Figure 3.5 Level 1 DFD ..... 18
Figure 3.6 Use case diagram ..... 19
Figure 3.7 Flowchart of the training process ..... 20
Figure 3.8 Flowchart of the scoring process ..... 21
Figure 3.9 State diagram of an applicant's decision ..... 21
Figure 3.16 score(): walking the tree ensemble to accumulate the raw margin and its … ..... 25
Figure 3.17 decision_for() and score_to_display(): the cutoff rule and the 300-850 sc… ..... 26
Figure 3.10 Cross-validated ROC-AUC of the candidate models ..... 27
Figure 3.11 Reliability diagram of the calibrated probabilities ..... 28
Figure 3.12 ROC curve of the calibrated model ..... 29
Figure 3.13 Feature importance of the twelve retained fields ..... 30
Figure 3.14 Observed versus predicted default rate by decile ..... 30
Figure 3.15 Approval rate, bad rate and cost per applicant against the cutoff ..... 31
Figure 3.18 Sequence diagram of a /api/predict-compare request ..... 32
Figure 3.19 score_forest(): averaging leaf probabilities across 300 trees instead of ad… ..... 33
Figure 4.1 Deployment of the Loan Default Risk Predictor ..... 43
Figure A-1 Blank applicant assessment form ..... 48
Figure A-2 A sample file loaded through a preset chip ..... 48
Figure A-3 A field-level validation error on submit ..... 49
Figure A-5 The landing page ..... 49
Figure A-6 A high-risk result slip with the score-drivers waterfall ..... 50
Figure A-7 The portfolio band split and cutoff explorer ..... 50
Figure A-8 The three-model comparison panel on the Assess page ..... 50
Figure A-9 The model benchmark on the methodology page ..... 51
Figure A-10 The assessment workbench on a mobile screen ..... 51

# LIST OF TABLES


Table 1.1 Comparison of existing approaches and the proposed system ..... 2
Table 2.1 Iterations followed during development ..... 7
Table 2.2 Actors and their responsibilities ..... 7
Table 2.3 Functional requirements ..... 8
Table 2.4 Role-permission matrix ..... 9
Table 2.5 Software used ..... 10
Table 2.6 Hardware used ..... 11
Table 3.1 Folder and file structure ..... 13
Table 3.2 Structure of the training dataset (fields used) ..... 15
Table 3.3 Encoding of categorical fields ..... 16
Table 3.4 Structure of model.json ..... 16
Table 3.5 Description of Level 1 processes ..... 18
Table 3.6 Use case description: Assess an applicant ..... 19
Table 3.7 Applicant, loan and credit position sections ..... 21
Table 3.8 Sample files and what-if controls ..... 22
Table 3.9 REST API endpoints ..... 23
Table 3.10 Prediction response fields ..... 24
Table 3.11 Cross-validated benchmark of candidate models ..... 27
Table 3.12 Search space and selected values ..... 27
Table 3.13 Out-of-fold evaluation metrics ..... 28
Table 3.14 Confusion matrix at the 0.25 cutoff ..... 29
Table 3.15 Approval, bad rate and cost by cutoff ..... 31
Table 3.16 Out-of-fold metrics of the three deployed models ..... 34
Table 4.1 Model fidelity and correctness tests ..... 37
Table 4.1A Multi-model deployment tests (added for the three-model rollout) ..... 37
Table 4.2 API validation and robustness tests ..... 38
Table 4.3 Model quality and performance tests ..... 39
Table 4.4 User interface tests ..... 39
Table 4.5 Endpoint and integration tests ..... 40
Table 4.6 Test summary ..... 41
Table 4.7 Defects found and corrections made ..... 42
Table 4.8 Using the system ..... 44

# LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| API | Application Programming Interface |
| AUC | Area Under the Curve |
| BCA | Bachelor of Computer Applications |
| CORS | Cross-Origin Resource Sharing |
| CSV | Comma-Separated Values |
| CV | Cross-Validation |
| DFD | Data Flow Diagram |
| DM | Deutsche Mark |
| ERD | Entity Relationship Diagram |
| FN | False Negative |
| FP | False Positive |
| GB | Gradient Boosting |
| JSON | JavaScript Object Notation |
| KS | Kolmogorov-Smirnov (statistic) |
| ML | Machine Learning |
| OOF | Out-of-Fold |
| PD | Probability of Default |
| PR-AUC | Precision-Recall Area Under the Curve |
| REST | Representational State Transfer |
| ROC | Receiver Operating Characteristic |
| STR | Summer Training Report |
| UCI | University of California, Irvine |
| UI | User Interface |
| XAI | Explainable Artificial Intelligence |

# C O N T E N T S


**CHAPTER 1: PROBLEM FORMULATION** ..... **1**

1.1 Introduction about the Company ..... 1

1.2 Introduction about the Problem ..... 1

1.3 Present State of the Art ..... 2

1.4 Need of Computerization ..... 3

1.5 Proposed Software / Project ..... 3

1.6 Importance of the Work ..... 4

**CHAPTER 2: SYSTEM ANALYSIS** ..... **6**

2.1 Feasibility Study ..... 6

2.2 Analysis Methodology ..... 7

2.3 Choice of the Platforms ..... 10

**CHAPTER 3: SYSTEM DESIGN** ..... **12**

3.1 Design Methodology ..... 12

3.2 Database Design ..... 14

3.3 Input Design ..... 21

3.4 Output Design ..... 22

3.5 Code Design and Development ..... 23

3.6 Model Design and Development ..... 26

**CHAPTER 4: TESTING AND IMPLEMENTATION** ..... **35**

4.1 Testing Methodology ..... 35

4.2 Test Data & Test Cases ..... 36

4.3 Test Reports and Debugging ..... 41

4.4 Implementation ..... 43

**CHAPTER 5: CONCLUSION AND REFERENCES** ..... **45**

5.1 Summary of the Work Done ..... 45

5.2 Conclusion ..... 45

5.3 Limitations of the System ..... 46

5.4 Future Scope for Modification ..... 46

5.5 References / Bibliography ..... 47

**CHAPTER 6: ANNEXURES** ..... **48**

A-1 Sample Inputs ..... 48

A-2 Sample Outputs ..... 49

A-3 Coding ..... 51

# CHAPTER 1

## PROBLEM FORMULATION

**Learning Objectives**

- Understand the training background and the company behind the internship
- Identify the problem lenders face in assessing credit risk
- Study the present state of the art and the need for computerisation
- Define the objectives, scope and importance of the Loan Default Risk Predictor

# CHAPTER 1: PROBLEM FORMULATION

## 1.1 Introduction about the Company

GRAStech is an organisation that runs online internship programs in technology fields. The program undertaken by the trainee was the six-week online internship program in Machine Learning Using Python. The program is headed by Mohammad Qayam Ali, Founder and Program Head.

The compulsory six-week summer training was completed as an online internship in Machine Learning Using Python at GRAStech, and the company issued a Certificate of Completion dated 31 July 2026, signed by Mohammad Qayam Ali. A copy of this certificate is placed after the Institute's Certificate at the beginning of this report. The training was carried out under the guidance of the college mentor, Ms. Nisha Choudhary, Assistant Professor. During the internship the trainee learned the complete life cycle of a machine learning application: data understanding, data preparation, modelling, evaluation and deployment, along with the testing that every stage requires. The project titled Loan Default Risk Predictor was designed and developed individually by the trainee as the training project.

## 1.2 Introduction about the Problem

A lender that approves an unsecured loan is taking on a risk that is settled only months or years later, when the applicant has either repaid the loan or defaulted on it. The decision to approve, decline or ask for conditions has to be made at the time of application, from the information the applicant has provided and the lender's own records, and it has to be consistent across every officer who makes it. Where that decision is left entirely to individual judgement, two applicants with a similar credit history can receive different outcomes depending on who reviewed the file, and the reasoning behind a decline is rarely written down in a form the applicant can be shown.

A default is costly to the lender in a way that a cautious decline is not. Every unit of loan that is extended to an applicant who later defaults has to be weighed against the much smaller cost of asking a genuinely creditworthy applicant to wait or to provide additional conditions, so the two kinds of mistakes are not equally expensive and a sound decision rule has to account for that difference rather than treat every error alike. Statistical scorecards have been used for decades to make this judgement more consistent, but a scorecard built on a simple weighted sum of the applicant's attributes does not capture the way several factors interact, for instance how a short employment history matters far more when it is combined with a long loan duration than on its own. Machine learning models that are flexible enough to capture such interactions exist, but most are used as a black box: the model returns a number, and neither the applicant nor the loan officer can see which factors pushed the score up or down, which makes the decision difficult to explain, to audit or to contest.

The problem can therefore be summarised in one statement: there is no simple, free and explainable tool that turns an applicant's details into a calibrated probability of default, a lending recommendation and a factor-by-factor reason for that recommendation, in a way a loan officer can defend and an applicant can understand.

### 1.3 Present State of the Art

Credit risk assessment today is carried out through a mix of approaches. Purely manual, expert-judgement underwriting relies on an officer's experience and a checklist of documents, and remains common for small lenders, but it does not scale and is hard to make consistent. Traditional logistic-regression scorecards, of the kind described by Siddiqi, assign points to binned attributes and sum them into a score; they are easy to explain because each bin contributes a fixed number of points, but they are built around a linear combination of the inputs and need manual binning to approximate non-linear effects. Credit bureau scores aggregate a borrower's history across institutions and are useful where that history exists, but they are an external, general-purpose number rather than something a lender can tune to its own cost of a bad loan. Black-box machine learning models, typically gradient boosting or random forest ensembles, are now used by many larger lenders because they handle non-linear and interaction effects well, but deployed without a calibration step or an explanation layer, their raw output is an uninterpreted score rather than a probability, and individual decisions cannot easily be justified.

A comparison of these approaches with the proposed system is given in Table 1.1.

**Table 1.1 Comparison of existing approaches and the proposed system**

| Capability | Manual underwriting | Traditional scorecard | Black-box ML | Proposed system |
|---|---|---|---|---|
| Probability output | No | Yes | Yes | Yes |
| Calibrated | No | Partial | Rarely | Yes |
| Explanation per applicant | Varies by officer | Yes | No | Yes |
| Handles non-linear effects | No | No | Yes | Yes |
| Cost-aware cutoff | No | Partial | Rarely | Yes |
| Deployment without a heavy runtime | Yes | Yes | No | Yes |
| Free and open | Varies | Varies | Varies | Yes |

The table shows that no single existing approach combines a calibrated probability, a per-applicant explanation, non-linear modelling and a cost-aware cutoff, all deployed without a heavy runtime and at no licensing cost. The Loan Default Risk Predictor is designed to occupy that space.

## 1.4 Need of Computerization

A computerised, explainable system is needed because the present mix of manual and black-box approaches suffers from the following drawbacks:

- Manual decisions are inconsistent, since two officers can read the same application differently.
- A plain score, from a scorecard or a black-box model, gives no reason the applicant or officer can inspect.
- Black-box models are rarely checked for calibration, so a quoted probability may not match the true default rate.
- A missed default and a wrongly declined applicant cost differently, and few processes make that trade-off explicit or adjustable.
- There is no easy way to ask "what would change if the cutoff moved", since re-running a model by hand for every cutoff is impractical.
- Portfolio-level questions, such as how the bad rate splits across bands, are hard to answer without a system that aggregates every scored application.

A computerised and explainable system removes these drawbacks: it applies the same calibrated model to every applicant, produces a probability and a factor-by-factor reason in a fraction of a second, lets the cutoff be moved and inspected instantly, and aggregates every prediction into the portfolio view a risk analyst needs.

## 1.5 Proposed Software / Project

The Loan Default Risk Predictor is a web application that scores a loan applicant's probability of default with a calibrated gradient boosting model and explains the score factor by factor. It is a three-tier application consisting of a Next.js and React front end, dependency-free Python scoring functions deployed as Vercel serverless functions (three models scored on request), and the JSON artifacts produced by an offline training pipeline. The live application is available at https://credoscore-gk.vercel.app.

### 1.5.1 Objectives of the Project

1. To collect the twelve applicant, loan and credit-position fields needed to score a loan application, with client-side and server-side validation.
2. To benchmark four candidate models, deploy three of them to production (gradient boosting, XGBoost and random forest), and score them together on every request for cross-verified agreement rather than one model's self-reported accuracy.
3. To return, for every scored applicant, a probability of default, a risk band, a credit-style score and a lending recommendation.

4. To explain every prediction with a factor-by-factor contribution that sums exactly to the predicted probability.
5. To let a user move the approval cutoff and see the resulting approval rate, bad rate among approved applicants and expected cost change in real time.
6. To provide portfolio-level analytics, namely the split of applicants by risk band, the cutoff trade-off, the decile-wise lift and the default rate by segment.
7. To deploy the system on a free hosting platform so that it can be accessed publicly through a web link, without requiring a heavy machine learning runtime at request time.

### 1.5.2 Scope of the Project

The scope of the project covers the following:

- Capturing the twelve fields used by the model and validating them on both the browser and the server.
- Loading one of several sample applicant files to fill the form quickly.
- Scoring an applicant and showing the probability of default, the risk band, the score and the recommendation.
- Explaining the score with a waterfall of individual factor contributions.
- A what-if control that recomputes the band, the score and the decision as the cutoff is moved.
- Portfolio analytics: band split, cutoff trade-off, deciles and segments.
- A methodology page describing the model, its benchmark against other candidate models, its calibration and its known limitations.
- A JSON REST API exposing the same scoring and portfolio functionality independently of the web interface.

The following are outside the present scope: using the tool for real lending decisions, a formal fairness audit of the model against protected characteristics, user accounts, storing submitted applications, and retraining the model from within the user interface. These are listed as future enhancements in Chapter 5.

## 1.6 Importance of the Work

The work addresses a problem at the centre of consumer lending, turning an applicant's details into a decision that is both accurate and explainable, grounded in a real, well-studied dataset rather than an invented scenario. The benchmarking, tuning, calibration and explanation method were all carried out by the trainee rather than taken untouched from an existing library, which makes the project useful as a teaching example of an explainable, cross-verified scoring system. It also gave practical

exposure to the full life cycle of an applied machine learning product, from feature engineering and model benchmarking through calibration, explanation, a dependency-free production API, a front end and testing to cloud deployment.

The project also demonstrates professional practices expected of a developer working on a credit model: the personal-status-and-sex attribute was excluded as a protected characteristic before any modelling began, the model's output was calibrated and evaluated honestly on out-of-fold predictions, no applicant data is retained by the stateless, secret-free deployed API, and the report states plainly, wherever the tool is shown, that it is not meant for real lending decisions.

# CHAPTER 2

## SYSTEM ANALYSIS

**Learning Objectives**

- Assess the feasibility of the project
- Describe the analysis methodology and the requirements
- Identify actors and permissions
- Justify the choice of software and hardware

# CHAPTER 2: SYSTEM ANALYSIS

## 2.1 Feasibility Study

Before development began, a feasibility study checked whether the proposed system could be built within the available time and skills, and whether it would be useful in practice.

### Technical Feasibility

All the technologies chosen are open source, well documented and widely used: Next.js and React for the front end, Python's standard library for the scoring functions, and pandas, scikit-learn and XGBoost for training. The trainee had the required background in JavaScript, TypeScript and Python, and received guidance for the modelling and calibration steps. The German Credit dataset trains in under a minute on an ordinary laptop, and Vercel's free tier auto-detects Next.js and deploys Python functions alongside it. The project is therefore technically feasible.

### Economic Feasibility

The cost of the system is effectively zero: the dataset is CC BY 4.0, every library used is open source, development ran on an ordinary laptop, and Vercel's free tier is sufficient for a project of this size. The benefit, a faster and more consistent credit decision, outweighs this negligible cost.

### Operational Feasibility

The interface follows the pattern of an ordinary web form: labelled fields, a submit button, a results panel. A loan officer needs no training to fill in twelve fields or read a probability, a band and a recommendation, and a risk analyst reads the portfolio pages like any dashboard, so operational acceptance is expected to be high.

### Schedule Feasibility

The six-week training period was divided into six weekly iterations, developed and tested one after another, as shown in Figure 2.1. The schedule was realistic and the project was completed within it.

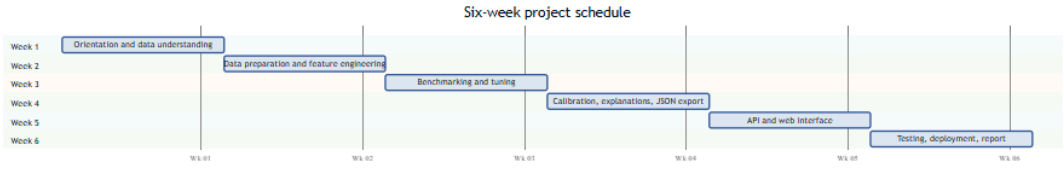



**Figure 2.1 Project schedule of the six-week training**

### Legal and Security Feasibility

The UCI Statlog German Credit Data is licensed CC BY 4.0 and attributed to Professor Hans Hofmann of Universität Hamburg, so its use here is legally clear. The personal-status-and-sex

attribute is excluded as a protected characteristic, and the deployed application stores no applicant data; every request is stateless. The report and the application state throughout that the tool is a student project, not meant for real lending decisions.

## 2.2 Analysis Methodology

An iterative, CRISP-DM-style approach was followed, in which data understanding, preparation, modelling, evaluation and deployment were revisited in successive iterations rather than attempted once end to end. Each iteration produced a working piece of the system, tested before the next began, as summarised in Table 2.1.

**Table 2.1 Iterations followed during development**

| Iteration | Work carried out | Result |
|---|---|---|
| 1 | Orientation and data understanding | The German Credit dataset, its attributes and its licence were studied, and the attributes to exclude were decided |
| 2 | Data preparation and feature engineering | Employment buckets, derived monthly income and one-hot encoding of categorical fields were implemented |
| 3 | Benchmarking and tuning | Four candidate models were compared under five-fold cross-validation and the gradient boosting model was tuned by randomized search |
| 4 | Calibration, explanation and export | Platt scaling, the Saabas contribution rescaling and the model.json / portfolio.json export were implemented |
| 5 | API and web interface | The Python scoring and portfolio functions and the Next.js front end were built |
| 6 | Testing, deployment and report | Notebook, API and interface tests were run, the application was deployed, and this report was written |

### 2.2.1 Requirement Gathering

Requirements were gathered by studying how credit scorecards and bureau scores are used in practice, working through the UCI dataset's own documentation to see what information is realistically available at the point of application, and reviewing the requirement with the training mentor before design started.

### 2.2.2 Users of the System

The system is used by three actors, whose responsibilities are set out in Table 2.2.

**Table 2.2 Actors and their responsibilities**

| Actor | Responsibility |
|---|---|
| Loan officer | Enters an applicant's details or loads a sample file, reads the probability of default, the score and the recommendation, reviews the factor-by-factor explanation, and moves the what-if cutoff for an individual applicant. |

| Actor | Responsibility |
|---|---|
| Risk analyst | Reads the portfolio-level analytics (band split, cutoff trade-off, deciles, segments) and the methodology page to judge whether the deployed cutoff and the model remain appropriate. |
| Model maintainer | Retrains the model on updated data, keeps the exported model.json and portfolio.json artifacts in sync with the decision thresholds in decision.ts, and is responsible for the model's version number. |

### 2.2.3 Functional Requirements

The requirements gathered from the dataset, the training mentor and the trainee's own study of credit scoring were consolidated into the fourteen functional requirements listed in Table 2.3.

**Table 2.3 Functional requirements**

| ID | Requirement | Actor |
|---|---|---|
| FR1 | The system shall capture the twelve required applicant, loan and credit-position fields, with client-side and server-side validation of numeric bounds, known categories and the instalment cross-field rule. | Loan officer |
| FR2 | The system shall allow a sample applicant file to be loaded to pre-fill the form. | Loan officer |
| FR3 | The system shall return, for every submitted applicant, a probability of default, a risk band, a credit-style score and a lending recommendation. | Loan officer |
| FR4 | The system shall show a factor-by-factor contribution waterfall whose values sum exactly to the predicted probability of default. | Loan officer |
| FR5 | The system shall allow the approval cutoff to be moved in a what-if control and shall recompute the band, score and decision accordingly. | Loan officer |
| FR6 | The system shall show a portfolio-level split of applicants by risk band. | Risk analyst |
| FR7 | The system shall let the user explore the approval rate, the bad rate among approved applicants and the expected cost across a grid of cutoffs. | Risk analyst |
| FR8 | The system shall show the model's decile-wise observed default rate against its predicted rate. | Risk analyst |
| FR9 | The system shall show the default rate across applicant segments. | Risk analyst |
| FR10 | The system shall provide a methodology page describing the model, its benchmark against other candidate models, its calibration and its known limitations. | Risk analyst |
| FR11 | The system shall allow the model maintainer to retrain the model and export it as a versioned JSON artifact. | Model maintainer |
| FR12 | The system shall show field-level validation errors on the form, with `aria-invalid` attributes and focus moved to the first invalid field. | Loan officer |

| ID | Requirement | Actor |
|---|---|---|
| FR13 | The system shall expose the scoring and portfolio functionality through a JSON REST API with CORS enabled, usable independently of the web interface. | Model maintainer |
| FR14 | The system shall provide a light and dark theme, selectable by the user, and a layout usable from 360px of viewport width upward. | Loan officer |

### 2.2.4 Non-Functional Requirements

- Security: strict server-side validation of every field, no stack traces returned to the client, a stateless API that stores no applicant data.
- Usability: field-level error messages, sample files to try the system quickly, a layout that works on a phone screen.
- Performance: the scoring function returns a result in well under a second; the underlying tree-walk and explanation computation measured a mean latency of 0.729 ms and a p95 of 0.907 ms over 1,000 calls.
- Reliability: invalid input must never crash the scoring function; every request returns a 2xx or a 400 response with a clear message, and there are no 500 errors.
- Portability: the front end works in every modern browser and on mobile screens, and the scoring function has no runtime dependency beyond the Python standard library.
- Maintainability: the decision thresholds are kept in one place on each side (decision.ts on the front end, predict.py on the back end) so that a change to the cutoff logic can be made and checked in both places together.
- Accessibility: labelled form fields, visible focus states, aria-invalid on failed fields, and colours for the risk bands chosen to be distinct from the brand colour and from each other.

### 2.2.5 Role-Permission Matrix

The permissions granted to each actor over every operation are set out in Table 2.4.

**Table 2.4 Role-permission matrix**

| Operation | Loan officer | Risk analyst | Model maintainer |
|---|---|---|---|
| Submit an applicant for scoring | Yes | Yes | Yes |
| View the score and the factor explanation | Yes | Yes | Yes |
| Adjust the what-if cutoff | Yes | Yes | Yes |
| View portfolio analytics | View only | Yes | Yes |
| View the methodology and limitations page | View only | Yes | Yes |
| Call the JSON API directly | No | No | Yes |
| Retrain and export the model | No | No | Yes |
| Change the decision thresholds | No | No | Yes |

## 2.3 Choice of the Platforms

The technologies were selected after comparing the alternatives against four criteria: cost, suitability for a small tabular dataset and an explainable model, fit with a stateless serverless deployment, and community support.

### 2.3.1 S/W used

The software stack chosen for the training pipeline, the API and the web interface is listed in Table 2.5.

**Table 2.5 Software used**

| Component | Technology | Reason for choice |
|---|---|---|
| Front-end framework | Next.js 16.3.5 (App Router, Turbopack) | Server-rendered React, file-based routing, fast dev server. |
| UI library | React 19.2.8 | Component model shared by the client islands. |
| Language | TypeScript 5 | Static typing catches shape errors in the scoring pipeline early. |
| Styling | Tailwind CSS 4 | Utility classes keep the editorial theme consistent. |
| UI primitives | Radix UI | Accessible, unstyled primitives for forms and overlays. |
| Charting | Recharts 3 | Declarative charts for the waterfall, band split and cutoff explorer. |
| Motion | Framer Motion | Restrained reveal and staircase-bar animations. |
| Icons | lucide-react | A consistent icon set for the navigation and form. |
| Scoring runtime | Python standard library (api/*.py) | Dependency-free handlers avoid shipping scikit-learn. |
| Data handling (training) | pandas 2.2.3, numpy 2.1.3 | Data loading and array computation during training. |
| Modelling (training) | scikit-learn 1.9.1, xgboost 2.1.3 | Candidate models under cross-validation; three refit and deployed. |
| Explanation cross-check | shap 0.46.0 | Independent check of the Saabas contributions used in production. |
| Training reporting | matplotlib 3.9.2, jupyter | Benchmarking, evaluation charts and test documentation. |
| Hosting | Vercel | Auto-detects Next.js, deploys api/*.py as Python functions. |

### 2.3.2 H/W used

The hardware used for development, and the minimum hardware expected at the client end, is listed in Table 2.6.

**Table 2.6 Hardware used**

| Component | Specification |
|---|---|
| Processor | Intel Core i5 or equivalent (or higher) |
| RAM | 8 GB minimum, 16 GB recommended |
| Storage | 256 GB SSD, at least 2 GB free |
| Display | Standard laptop or desktop monitor (1366 x 768 or higher) |
| Network | Broadband internet connection |
| Client device (end user) | Any computer, tablet or smartphone with a modern web browser |

### 2.3.3 Software Requirement Summary

To run the project locally, Node.js, Python 3.12, npm and a code editor are needed, with the training dependencies (pandas, numpy, scikit-learn, xgboost, shap, matplotlib, jupyter) if the model is to be retrained. Development runs three local Python servers alongside the Next.js application, which rewrites requests to them in development only. For the end user, only a browser and an internet connection are required.

# CHAPTER 3

## SYSTEM DESIGN

**Learning Objectives**

- Explain the system's architecture, module hierarchy and project structure
- Design the data structures and process models: ERD, DFD, use case diagram and flowcharts
- Describe the input design, the output design, the REST API and the security design
- Describe how the scoring model was selected, tuned, calibrated, evaluated and explained

# CHAPTER 3: SYSTEM DESIGN

## 3.1 Design Methodology

The system was designed top-down: architecture, module boundaries, folder structure, data design, then screens. Structured design tools were used throughout: an ERD for the data the system reads and writes, DFDs for the movement of a request through the scoring pipeline, a use case diagram for actor-system interaction, and flowcharts and a state diagram for training and scoring.

### 3.1.1 System Architecture

The Loan Default Risk Predictor follows a three-tier architecture, shown in Figure 3.1. The browser never evaluates a model itself; every scoring request goes to a server-side function, and every reusable number used to build a page is read from a JSON artifact at build time.

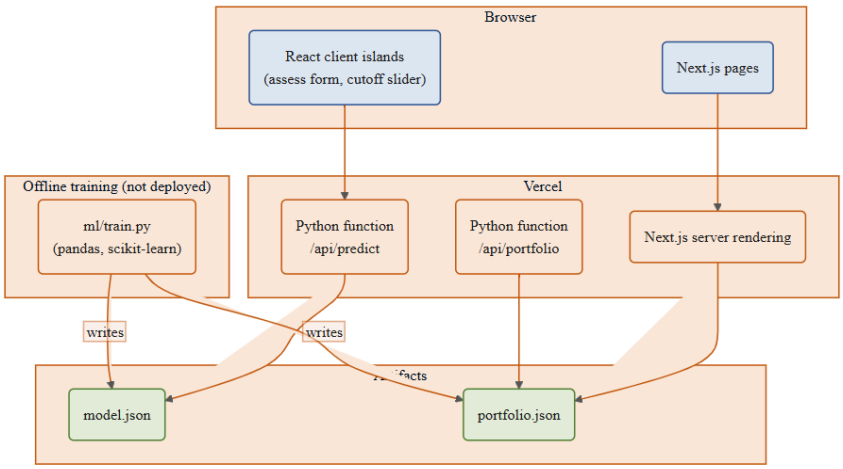



**Figure 3.1 Three-tier architecture of the Loan Default Risk Predictor**

- **Presentation layer.** Next.js 16 App Router pages (Brief, Assess, The book, Methodology) rendered on the server and hydrated as React 19 client islands only where interaction is required, for example the applicant form and the cutoff slider.
- **Application layer.** Vercel hosts the Next.js server and three standard-library Python functions, `api/predict.py`, `api/predict-compare.py` and `api/portfolio.py`, the only parts of the system the browser calls over the network at runtime.
- **Data layer.** Three JSON artifacts, `model/model.json`, `model/model_xgboost.json`, `model/model_random_forest.json`, plus `model/portfolio.json`, hold the three fitted ensembles and the precomputed portfolio aggregates. A separate offline path, `ml/train.py`, reads the raw dataset with pandas and scikit-learn and writes all four; it never runs in production.

### 3.1.2 Module Hierarchy

The system divides into four modules, shown in Figure 3.2: the training pipeline, the scoring API, the web interface and the test suite. The training pipeline prepares data, benchmarks candidate models, tunes and calibrates gradient boosting, exports all three deployed artifacts. The scoring API validates a request, walks the relevant ensemble(s), calibrates and bands the result, explains it where applicable and serves the portfolio analytics. The web interface covers the four pages reachable from the navigation bar. The test suite is the notebook's model tests together with the API and UI test scripts described in Chapter 4.

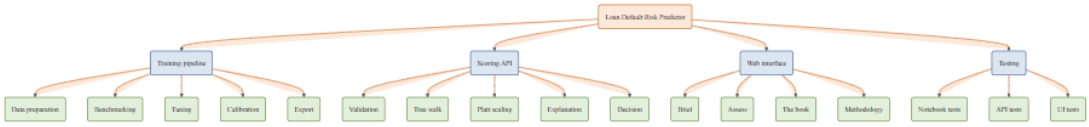


**Figure 3.2 Module hierarchy of the Loan Default Risk Predictor**

### 3.1.3 Project Structure

The repository's folder and file structure is summarised in Table 3.1, verified against the project tree after the user interface was rebuilt.

**Table 3.1 Folder and file structure**

| File / Folder | Purpose |
|---|---|
| `app/layout.tsx`, `app/page.tsx`, `app/globals.css` | Root layout, Brief landing page, global stylesheet. |
| `app/assess/`, `app/portfolio/`, `app/model/` | The Assess, The book and Methodology pages. |
| `app/error.tsx`, `app/not-found.tsx` | Route-level error and 404 screens. |
| `components/assess/` | Applicant form, workbench, waterfall, cutoff control, PD meter, verdict stamp, model comparison, presets. |
| `components/portfolio/` | Risk-band split, cutoff explorer, decile chart/table, segment explorer. |
| `components/model/` | Benchmark table, calibration chart, cutoff table, importance bars, ROC chart. |
| `components/landing/` | Calibration miniature, cost-asymmetry panel, decile staircase, decision pipeline diagram. |
| `components/site/`, `components/editorial/` | Nav, footer, theme toggle; shared layout primitives (section, stat, figure, reveal). |
| `components/ui/`, `components/charts/chart-kit.tsx` | Radix-based primitives and the shared Recharts configuration. |

| File / Folder | Purpose |
|---|---|
| `lib/types.ts`, `lib/format.ts`, `lib/utils.ts` | Shared types, label/number formatters, the `cn()` helper. |
| `lib/decision.ts`, `lib/portfolio.ts`, `lib/model-meta.ts` | Client decision mirror, portfolio reader, server-only model metadata reader. |
| `lib/api.ts` | The typed `predict()`/`predictCompare()` clients for the scoring endpoints. |
| `api/predict.py`, `api/predict-compare.py`, `api/portfolio.py` | The three Python functions deployed as Vercel serverless functions. |
| `ml/train.py`, `ml/dev_server.py` | Offline training pipeline; local dev server mirroring the three functions on ports 8001/8002/8003. |
| `model/model.json`, `model/model_xgboost.json`, `model/model_random_forest.json`, `model/portfolio.json` | Exported ensembles (gradient boosting, XGBoost, random forest) and precomputed portfolio aggregates. |
| `data/german_credit.csv` | The cleaned, re-labelled training dataset. |
| `notebook.ipynb` | Reproducibility and quality-gate tests (Chapter 4). |
| `docs/` | Internal engineering notes (architecture, data, model, testing, deployment, limitations). |
| `vercel.json`, `next.config.ts` | Function bundling config and the local rewrite to the three Python servers. |

## 3.2 Database Design

The system has no transactional database. Its data design covers the training dataset and the JSON artifacts the API reads.

### 3.2.1 ERD

An applicant submits one loan application, which carries exactly one credit position and receives exactly one prediction per deployed model; a prediction is explained by one or more per-factor contributions. The relationships are drawn in Figure 3.3.

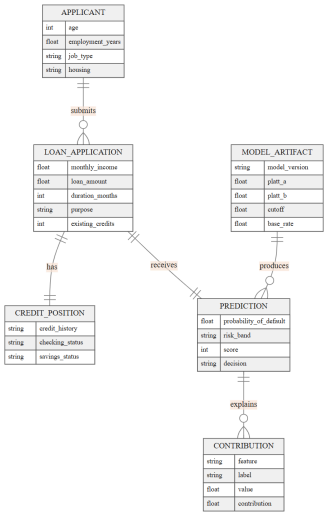



**Figure 3.3 Entity Relationship Diagram of the Loan Default Risk Predictor**

### Collection Structure

The training dataset's twelve retained fields are structured as shown in Table 3.2.

**Table 3.2 Structure of the training dataset (fields used)**

| Field | Type | Values / range | Source attribute |
|---|---|---|---|
| age | Numeric | 18–100 years | Attribute 13 |
| employment_years | Numeric | 0, 0.5, 2.5, 5.5, 10 (bucket midpoints) | Attribute 7 (A71–A75) |
| monthly_income | Numeric | 1–1,000,000 DM | Engineered from Attributes 2, 5 and 8 |
| loan_amount | Numeric | 1–1,000,000 DM | Attribute 5 |
| duration_months | Numeric | 1–120 months | Attribute 2 |
| existing_credits | Numeric | 0–20 | Attribute 16 |
| job_type | Categorical | management, skilled, unskilled_nonresident, unskilled_resident | Attribute 17 |
| housing | Categorical | free, own, rent | Attribute 15 |
| purpose | Categorical | business, car_new, car_used, domestic_appliances, education, furniture_equipment, other, radio_tv, repairs, retraining | Attribute 4 |
| credit_history | Categorical | all_paid, critical, delayed_past, no_credits, paid_duly | Attribute 3 |
| checking_status | Categorical | 0_to_200, 200_or_more, less_than_0, none | Attribute 1 |
| savings_status | Categorical | 100_to_500, 500_to_1000, none, over_1000, under_100 | Attribute 6 |

monthly_income does not exist as a UCI attribute; it is engineered at load time as `(loan_amount / duration_months) / (installment_rate / 100)`, so a 1,169 DM loan over 6 months at an installment rate of 4% implies a monthly income of 4,870.83 DM. Attribute 9 (personal status and sex) is excluded as a protected attribute; Attributes 10, 11, 12, 14, 18, 19 and 20 are excluded to keep the applicant form short.

Each of the six categorical fields is one-hot encoded; Table 3.3 gives the resulting column counts.

**Table 3.3 Encoding of categorical fields**

| Field | Categories | One-hot columns |
|---|---|---|
| job_type | 4 | 4 |
| housing | 3 | 3 |
| purpose | 10 | 10 |
| credit_history | 5 | 5 |
| checking_status | 4 | 4 |
| savings_status | 5 | 5 |

The six numeric fields pass through unscaled, so the full feature vector that the ensemble sees has 6 + 31 = 37 columns. The A47 vacation purpose code is absent from the data entirely, and A48 (retraining) is present in only 9 of the 1,000 rows.

`model/model.json`, the deployed artifact, is structured as shown in Table 3.4.

**Table 3.4 Structure of model.json**

| Key | Type | Description |
|---|---|---|
| model_version | String | Semantic version of the exported artifact ("1.0.0"). |
| model_type | String | "gradient_boosting". |
| trained_on_rows | Integer | 1,000. |
| numeric_features, categorical_features | Array of String | The six numeric and six categorical field names, in vector order. |
| categories | Object | Category list per categorical field, in encoding order. |
| learning_rate | Number | 0.05. |
| base_value | Number | The ensemble's starting log-odds, ln(0.3 / 0.7). |
| platt_a, platt_b | Number | The Platt scaling coefficients fitted on out-of-fold margins. |
| trees | Array of Object | 250 trees, each with left, right, feature, threshold and value arrays. |
| base_rate | Number | 0.30, the dataset's observed default rate. |
| cutoff | Number | 0.25, the deployed approval cutoff. |

| Key | Type | Description |
|---|---|---|
| metrics, benchmark | Object | Out-of-fold evaluation metrics and the cross-validated model comparison. |
| hyperparameters | Object | The selected values from the randomised search. |

**Validations, Verifications and Normalization**

- Field-level validation: the API checks that all twelve fields are present, that the six numeric fields are finite numbers within their documented bounds, and that the six categorical fields hold a value from the trained category list; any failure is rejected before scoring.
- Cross-field validation: the implied monthly instalment, loan_amount divided by duration_months, may not exceed four times monthly_income.
- Parity check: at export time, the hand-written tree walk in `api/predict.py` is compared against the fitted scikit-learn pipeline's output, and the exported artifact is accepted only when the maximum difference across the training rows is below 1e-9.
- Additivity check: the per-factor contributions produced during explanation are summed and compared against the probability minus the base rate; the notebook asserts this difference stays below 1e-14.
- Encoding normalisation: an unseen category in the request is rejected outright rather than silently mapped to a default, since the one-hot scheme has no "unknown" column.
- Privacy: the protected attribute (personal status and sex) is excluded from training altogether, and no applicant's input or output is persisted anywhere; the API is stateless.

### 3.2.2 DFD

Figure 3.4 is the context-level diagram (Level 0), which treats scoring and analytics as a single process bounded by three external entities. Figure 3.5 breaks that process into six processes at Level 1.

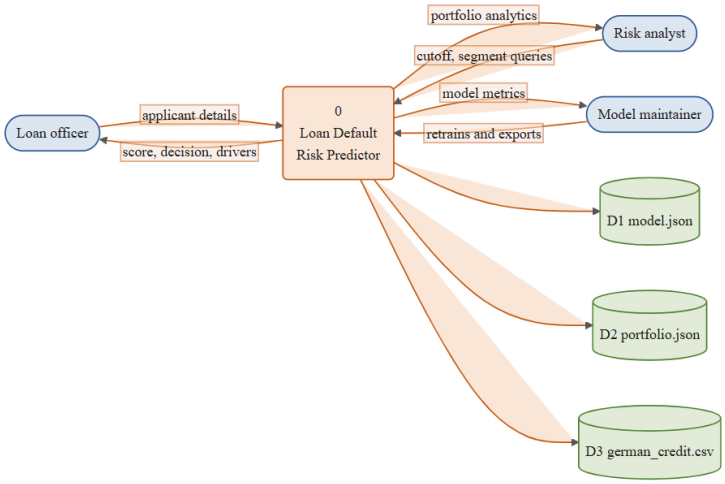



**Figure 3.4 Context Level DFD (Level 0)**

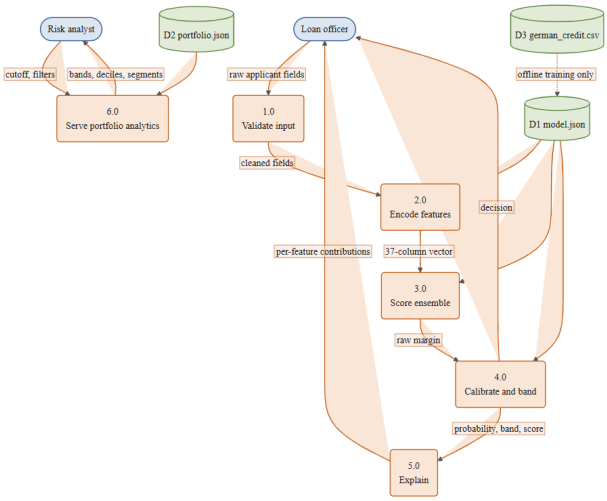



**Figure 3.5 Level 1 DFD**

**Table 3.5 Description of Level 1 processes**

| No. | Process | Description |
|---|---|---|
| 1.0 | Validate input | Checks that all twelve fields are present, numeric fields are in bounds, categorical fields are known, and the instalment-to-income rule holds; rejects with a field-level 400 otherwise. |
| 2.0 | Encode features | Builds the 37-column vector from the validated fields, in the exact numeric-then-categorical order recorded in D1. |
| 3.0 | Score ensemble | Walks all 250 trees of the gradient boosting ensemble, accumulating a raw log-odds margin and a per-feature path contribution. |
| 4.0 | Calibrate & band | Applies Platt scaling to the raw margin to obtain a probability of default, then assigns a risk band and a decision using the stored cutoff. |

| No. | Process | Description |
|---|---|---|
| 5.0 | Explain | Groups the per-feature contributions by source field and rescales them so they sum exactly to the probability minus the base rate. |
| 6.0 | Serve portfolio analytics | Reads D2 and returns the precomputed cutoff curve, deciles, calibration bins, feature importance and segments without any further computation. |

Data stores: D1 model.json (the fitted ensemble and its metadata), D2 portfolio.json (the precomputed, out-of-fold portfolio aggregates), and D3 german_credit.csv (the training dataset, read only by the offline training pipeline).

**Use Case Diagram**

Three actors use the system: a loan officer, who scores individual applicants; a risk analyst, who studies the portfolio-level analytics; and a model maintainer, who retrains and redeploys the artifact. Their use cases are shown in Figure 3.6.

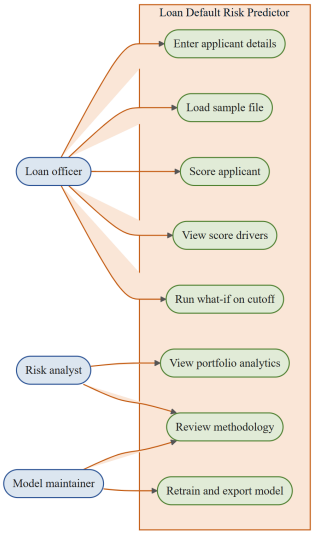



**Figure 3.6 Use case diagram**

**Table 3.6 Use case description: Assess an applicant**

| Item | Description |
|---|---|
| Actor | Loan officer |
| Pre-condition | The officer has opened the Assess page and has the applicant's twelve field values, or chooses a sample file instead. |
| Main flow | 1. The officer fills in the applicant form, or loads a sample file. 2. The officer submits the form. 3. The browser posts the twelve fields to `/api/predict`. 4. The server validates, scores, calibrates and explains the result. 5. The server returns the probability of default, the risk band, the score, the recommended decision and the per-factor contributions. 6. The browser renders the risk slip, the PD meter and the contribution waterfall. |

| Item | Description |
|---|---|
| Alternate flow | Validation fails: the offending field is highlighted with an inline message and focus moves to it, and nothing is submitted. Scoring service unreachable: a friendly error replaces the result panel and the form stays filled in so the officer can retry. |
| Post-condition | A risk slip with the probability of default, the band, the score and the decision is shown, and the officer may move the what-if cutoff slider to see how the recommendation would change. |

**Process Flowcharts**

Figure 3.7 is the flowchart of the offline training pipeline. Figure 3.8 is the flowchart of a single scoring request. Figure 3.9 is a state diagram of the decision an applicant's probability of default is mapped onto at the deployed cutoff.

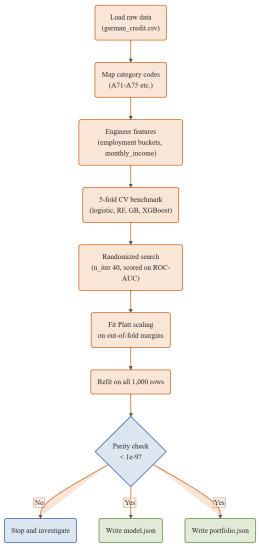



**Figure 3.7 Flowchart of the training process**

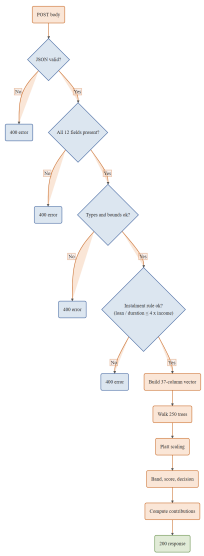


**Figure 3.8 Flowchart of the scoring process**

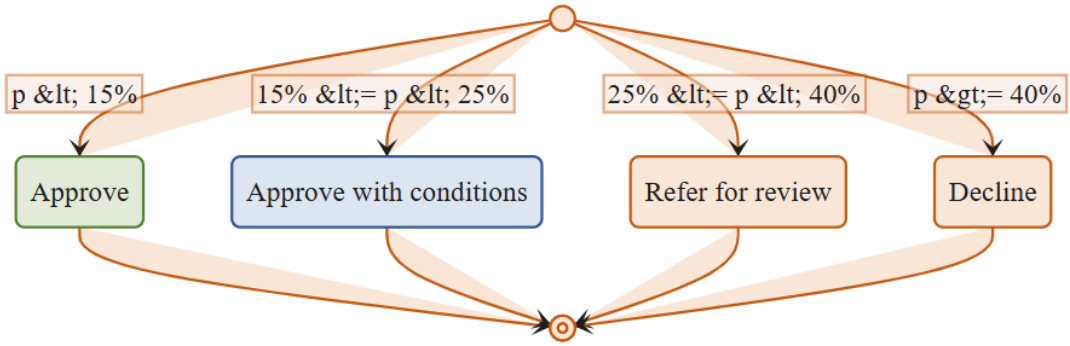



**Figure 3.9 State diagram of an applicant's decision**

## 3.3 Input Design

All data enters the system through the applicant form on the Assess page, which is divided into three sections. Required fields and bounded number inputs give the officer quick feedback in the browser, and every rule is checked again on the server, since the browser's checks can be bypassed by calling the API directly.

**Table 3.7 Applicant, loan and credit position sections**

| Section | Field | Control | Validation |
|---|---|---|---|
| Applicant | Age | Number box (years) | Required; 18–100 |
| Applicant | Years employed | Number box (years, 0.5 step) | Required; 0–50 |

| Section | Field | Control | Validation |
|---|---|---|---|
| Applicant | Job type | Drop-down | Required; one of four trained categories |
| Applicant | Housing | Drop-down | Required; one of three trained categories |
| Applicant | Monthly income | Number box (DM) | Required; 1–1,000,000 |
| Loan | Amount | Number box (DM) | Required; 1–1,000,000 |
| Loan | Duration | Number box (months) | Required; 1–120 |
| Loan | Purpose | Drop-down | Required; one of ten trained categories |
| Credit position | Credit history | Drop-down | Required; one of five trained categories |
| Credit position | Checking account | Drop-down | Required; one of four trained categories |
| Credit position | Savings account | Drop-down | Required; one of five trained categories |
| Credit position | Existing credit lines | Number box | Required; 0–20 |

A cross-field rule runs after the four sections: the implied monthly instalment (amount divided by duration) may not exceed four times the monthly income, and the error is attached to the amount field when it does.

**Table 3.8 Sample files and what-if controls**

| Field | Control | Values |
|---|---|---|
| Sample file | Toggle buttons | Typical file, Strong file, Stretched file; each loads a complete, realistic set of the twelve fields |
| What-if cutoff | Slider | 5%–60%, 1-point steps, starting at the deployed cutoff (25%) |

## 3.4 Output Design

Output is presented as a risk slip and a set of linked charts rather than a single number. The following outputs are described below, and the corresponding screenshots are attached in Annexure A-2.

- The risk slip: the probability of default, the risk band, the 300–850 score and the recommended decision (approve, approve with conditions, refer for review, decline), shown together as soon as a score comes back.
- The PD meter: the probability of default drawn on a 0–100% scale against the low/medium/high band edges.

- The decision stamp: the recommended decision rendered in the band's colour, never in the brand green, so that "approved" and "on-brand" are never confused.
- The contribution waterfall: the twelve factors' contributions to the probability, drawn on a diverging percentage-point scale so that factors pushing the score down and up are both legible.
- The what-if cutoff control: moves the approval cutoff and recomputes only the decision label, leaving the probability of default untouched.
- The risk-band split, the cutoff explorer, the decile chart and table, and the segment explorer, all on the portfolio page, each reading the precomputed aggregates in `model/portfolio.json`.
- The methodology page's benchmark table, ROC chart, calibration chart and importance bars, each explained in §3.6.
- Error messages, shown inline next to the offending field or as a form-level banner, for example "Enter a value from 18 to 100." and "The monthly instalment (amount ÷ months) can't be more than four times monthly income."

## 3.5 Code Design and Development

### 3.5.1 REST API Design

The system exposes three read endpoints, each implemented as a standard-library Python function, listed in Table 3.9.

**Table 3.9 REST API endpoints**

| Verb | Endpoint | Access | Purpose |
|---|---|---|---|
| POST | /api/predict | Public | Score one applicant against the deployed gradient boosting model and return the probability, band, score, decision and contributions |
| OPTIONS | /api/predict | Public | CORS preflight; returns 204 |
| POST | /api/predict-compare | Public | Score one applicant against all three deployed models (gradient boosting, XGBoost, random forest) and return all three results in one response, so the interface can show whether they agree |
| OPTIONS | /api/predict-compare | Public | CORS preflight; returns 204 |
| GET | /api/portfolio | Public | Return the precomputed portfolio aggregates, cached for one hour |
| OPTIONS | /api/portfolio | Public | CORS preflight; returns 204 |

The prediction response's fields are listed in Table 3.10.

**Table 3.10 Prediction response fields**

| Field | Type | Description |
|---|---|---|
| probability_of_default | Number | The calibrated probability, rounded to four decimal places. |
| risk_band | String | "low", "medium" or "high". |
| score | Integer | 850 minus 550 times the probability, rounded. |
| decision | String | "approve", "approve_with_conditions", "refer_for_review" or "decline". |
| cutoff | Number | The deployed approval cutoff used for the decision. |
| contributions | Array of Object | One entry per source field: feature, label, value and contribution, sorted by the size of the contribution. |
| base_rate | Number | The dataset's observed default rate (0.30). |
| model_version | String | The deployed artifact's version. |

### 3.5.2 Security Design

Security was treated as a design requirement from the start, even though the system holds no user accounts and no personal data:

- Strict input validation: every field is checked for presence, type and range before anything else runs, and an unseen category is rejected rather than coerced.
- A single error shape: every rejected request returns HTTP 400 with `{error, message}` and no stack trace; the handler guards the whole body in a try/except, so there are no 500s for bad input.
- Statelessness: nothing an applicant submits is written anywhere; each request is scored and forgotten.
- No secrets: the deployed functions need no API key, database password or signing secret.
- Open CORS by design: `Access-Control-Allow-Origin: *` is set on every endpoint, acceptable because every endpoint is read-only, stateless and free of personal data.
- The protected attribute (personal status and sex) is excluded from the model, so it cannot leak into a decision even indirectly.
- Deployment hygiene: `vercel.json`'s `excludeFiles` keeps the dataset, the notebook and the documentation out of the deployed function bundle.
- Validation is mirrored, not shared: the same bounds and instalment rule are enforced independently in `assess-form.tsx` and `api/predict.py`, so a request can never bypass the rule by skipping the browser.

### 3.5.3 Important Code Segments

Scoring loop (`api/predict.py`). Each tree's path contribution is attributed from a virtual root value of 0, not the tree's real root value, which is what makes the contributions exactly additive across the whole ensemble. The same function now also decides the split direction per model type, since gradient boosting and XGBoost do not agree on how a tie at the threshold is routed (§3.6.8 explains why).

```python
def score(vector: list[float], model: dict[str, Any]) -> tuple[float, dict[str, float]]:
    # sklearn routes a tie left on "<="; XGBoost's histogram splitter places
    # thresholds exactly on real values, so it needs a strict "<" instead.
    strict_less = model.get("model_type") == "xgboost"
    contributions: dict[int, float] = {}
    raw = model["base_value"]

    for tree in model["trees"]:
        node = 0
        attributed_value = 0.0
        while tree["left"][node] != -1:
            feature_idx = tree["feature"][node]
            threshold = tree["threshold"][node]
            go_left = (
                vector[feature_idx] < threshold
                if strict_less
                else vector[feature_idx] <= threshold
            )
            next_node = tree["left"][node] if go_left else tree["right"][node]
            next_value = tree["value"][next_node]

            delta = (next_value - attributed_value) * model["learning_rate"]
            contributions[feature_idx] = contributions.get(feature_idx, 0.0) + delta

            node = next_node
            attributed_value = next_value

        raw += attributed_value * model["learning_rate"]

    return raw, contributions
```

**Figure 3.16 score(): walking the tree ensemble to accumulate the raw margin and its factor contributions**

Platt calibration and the decision rule (`api/predict.py`). The raw margin is passed through a logistic sigmoid with the fitted A and B coefficients, and the calibrated probability is compared against the cutoff at three multipliers to choose a decision:

```python
def decision_for(probability: float, cutoff: float) -> str:
    if probability < cutoff * 0.6:
        return "approve"
    if probability < cutoff:
        return "approve_with_conditions"
    if probability < cutoff * 1.6:
        return "refer_for_review"
    return "decline"


def score_to_display(probability: float) -> int:
    # Maps [0, 1] probability of default onto a familiar 300-850 band,
    # inverted so a lower default probability yields a higher score.
    return round(850 - probability * (850 - 300))
```

**Figure 3.17 decision_for() and score_to_display(): the cutoff rule and the 300-850 score mapping**

The raw, log-odds-space contributions are rescaled afterwards into probability space so they sum exactly to the probability minus the base-rate probability, and the same bounds and instalment rule are checked a second time in the browser, in `components/assess/assess-form.tsx`, so the officer sees an inline error instead of waiting on a round trip. Both excerpts, along with the training pipeline's tree export and the Platt fit, are given in Annexure A-3.

## 3.6 Model Design and Development

### 3.6.1 Data Preparation

The training pipeline, `ml/train.py`, loads the raw UCI Statlog (German Credit Data) file, 1,000 rows with 700 good and 300 bad accounts, and keeps twelve of its twenty attributes, dropping personal status and sex as a protected attribute and six further attributes to keep the applicant form short. The employment attribute's five coded buckets (A71–A75) are mapped to numeric midpoints (0, 0.5, 2.5, 5.5 and 10 years), and monthly income is engineered from the loan amount, the duration and the instalment rate, since the dataset carries no income field of its own. The twelve retained fields are then split into six numeric columns, passed through unscaled, and six categorical columns, one-hot encoded into 31 columns, for a 37-column feature vector overall.

### 3.6.2 Model Selection

Four candidate models were benchmarked with five-fold stratified cross-validation, using identical folds (`StratifiedKFold(5, shuffle=True, random_state=42)`) for every model so the comparison is fair. A manual per-fold loop was used instead of scikit-learn's scoring helpers,

because the XGBoost scikit-learn wrapper raised a shape error against scikit-learn 1.9's string-named scorers. The results are given in Table 3.11 and plotted in Figure 3.10.

**Table 3.11 Cross-validated benchmark of candidate models**

| Model | ROC-AUC (± std) | PR-AUC |
|---|---|---|
| Logistic regression | 0.7737 ± 0.0212 | 0.5757 |
| Random forest (300 trees, depth 6) | 0.7883 ± 0.0277 | 0.6329 |
| Gradient boosting (150 trees, depth 3, lr 0.05) | 0.7859 ± 0.0188 | 0.6243 |
| XGBoost (200 trees, depth 3, lr 0.05, scale_pos_weight 700/300) | 0.7893 ± 0.0193 | 0.6299 |
| Gradient boosting, tuned | 0.7965 | n/a |

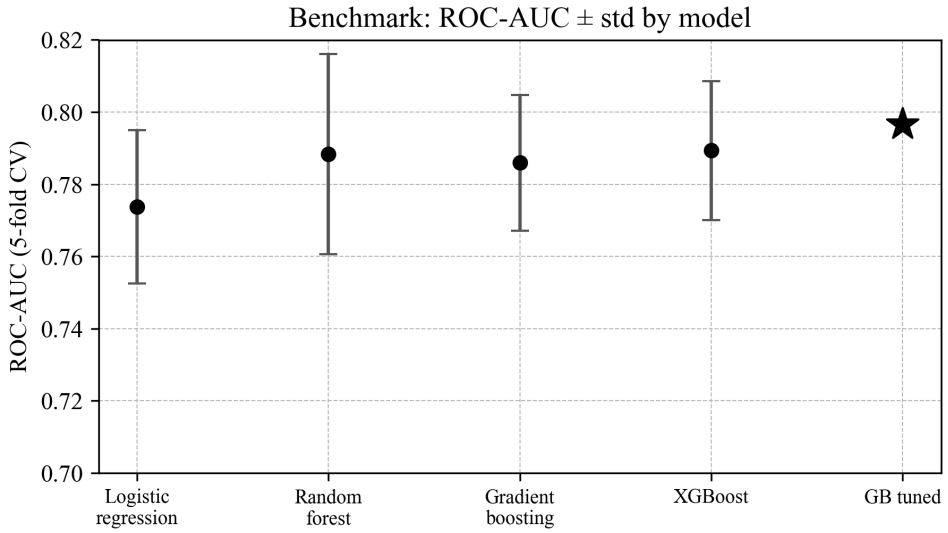



**Figure 3.10 Cross-validated ROC-AUC of the candidate models**

Gradient boosting was chosen over the marginally stronger untuned XGBoost and random forest results because, once tuned, it outperformed every other candidate, it needs no third-party package at inference time (the exported trees are walked by hand in pure Python), and that absence of a runtime dependency is what lets the scoring function deploy on Vercel without scikit-learn's roughly 100 MB footprint.

### 3.6.3 Hyperparameter Tuning

The chosen model was tuned with `RandomizedSearchCV`, 40 iterations, scored on ROC-AUC over the same five folds. Table 3.12 gives the search space and the values selected.

**Table 3.12 Search space and selected values**

| Hyperparameter | Search space | Selected value |
|---|---|---|
| n_estimators | 80, 120, 150, 200, 250, 300 | 250 |
| max_depth | 2–5 | 3 |

| Hyperparameter | Search space | Selected value |
|---|---|---|
| learning_rate | 0.01, 0.03, 0.05, 0.08, 0.1, 0.15 | 0.05 |
| subsample | 0.7–1.0 | 0.8 |
| min_samples_leaf | 5, 10, 20, 30 | 20 |

### 3.6.4 Probability Calibration

A gradient boosting ensemble's raw margin is a log-odds score, not a probability, and an uncalibrated score tends to be overconfident at the extremes. Platt scaling was fitted on the out-of-fold margins produced by `cross_val_predict`, giving $P = \sigma(A \cdot \text{raw} + B)$ with $A = 0.796129$ and $B = -0.113198$. At the ensemble's own starting log-odds, base_value $= \ln(0.3 / 0.7) = -0.847298$, this gives a model baseline probability of default of $\sigma(A \cdot \text{base_value} + B) \approx 0.3126$, close to the dataset's own 0.30 base rate, as Figure 3.11 confirms across the full probability range.

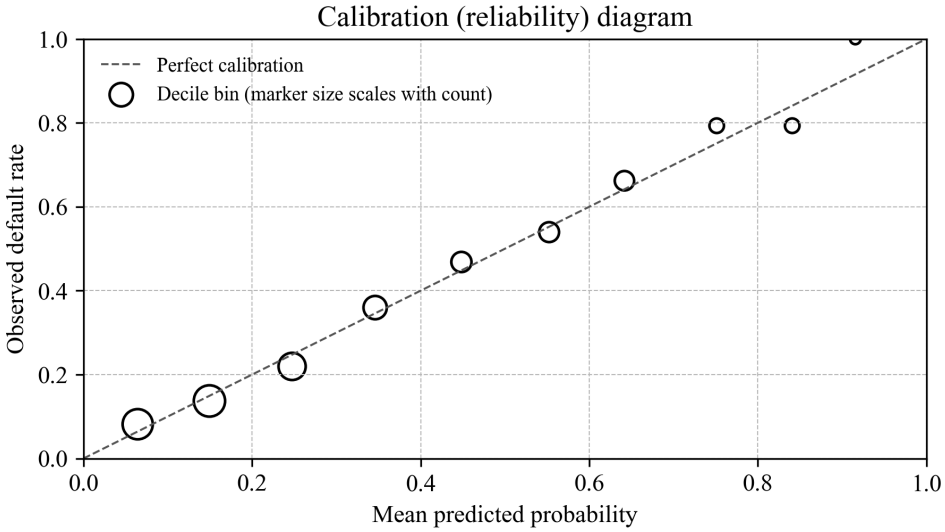



**Figure 3.11 Reliability diagram of the calibrated probabilities**

### 3.6.5 Evaluation

Every metric below is computed out-of-fold, so no applicant's score is evaluated on a model that was trained on that applicant's row. The headline metrics are given in Table 3.13.

**Table 3.13 Out-of-fold evaluation metrics**

| Metric | Value |
|---|---|
| ROC-AUC | 0.7928 |
| PR-AUC | 0.6344 |
| KS statistic | 0.4776 |
| Brier score | 0.1601 (flat-guess baseline 0.21) |
| Gini coefficient | 0.5855 |
| Weighted calibration gap | 0.0195 |

| Metric | Value |
|---|---|
| Top-decile vs bottom-decile lift | 10.9× |
| Band counts | low 327 / medium 336 / high 337 |

The ROC curve behind the ROC-AUC figure is plotted in Figure 3.12.

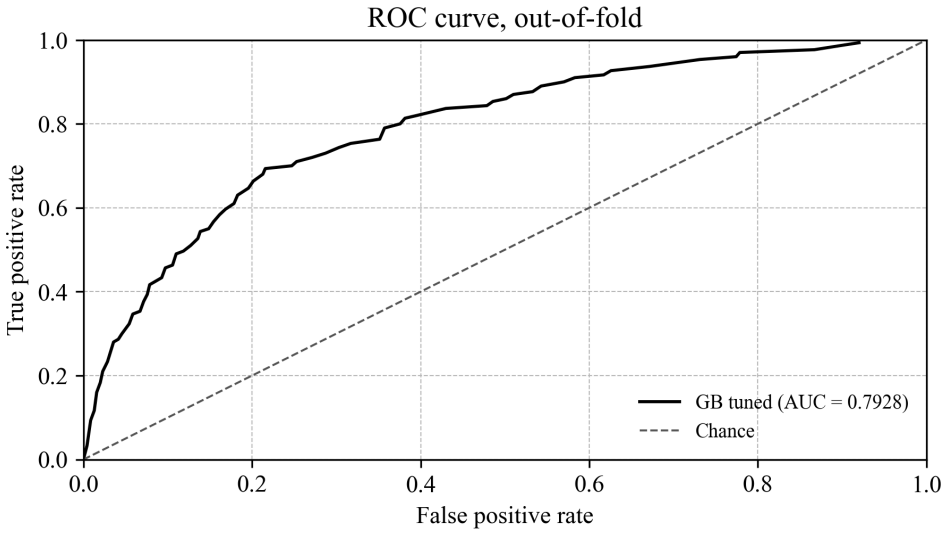



**Figure 3.12 ROC curve of the calibrated model**

At the deployed 0.25 cutoff, the applicants split into the confusion matrix in Table 3.14, where a predicted default is a declined application.

**Table 3.14 Confusion matrix at the 0.25 cutoff**

| | Predicted: Decline | Predicted: Approve |
|---|---|---|
| Actual: Default | TP = 233 | FN = 67 |
| Actual: No default | FP = 250 | TN = 450 |

This gives an accuracy of 0.683, a precision of 0.482, a recall of 0.777 and a specificity of 0.643 at a cost (five per missed default, one per wrongful decline) of 585, or 0.585 per applicant. checking_status, monthly_income and loan_amount drive most of this discrimination, as Figure 3.13's feature importance bars show, and the model's rank ordering holds up decile by decile: the top decile's observed default rate is 10.9 times the bottom decile's, as Figure 3.14 shows.

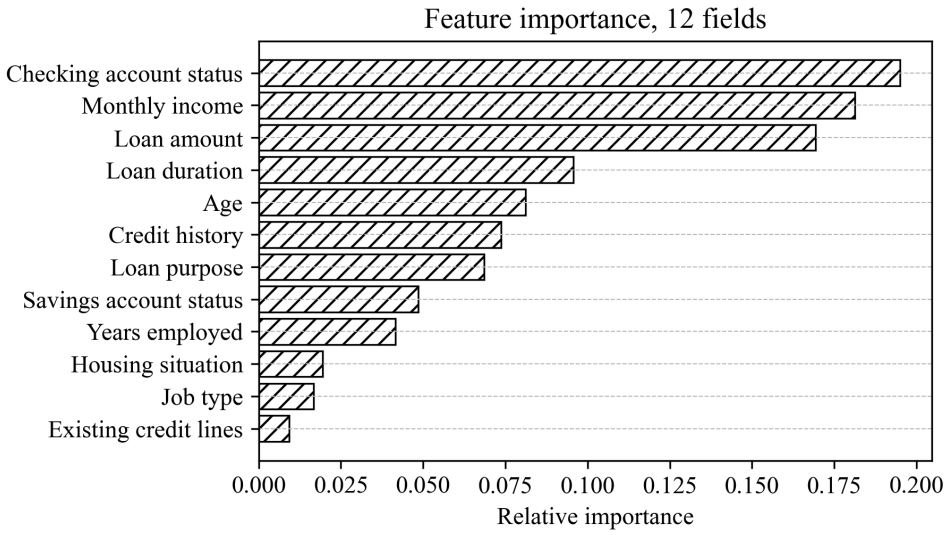



**Figure 3.13 Feature importance of the twelve retained fields**

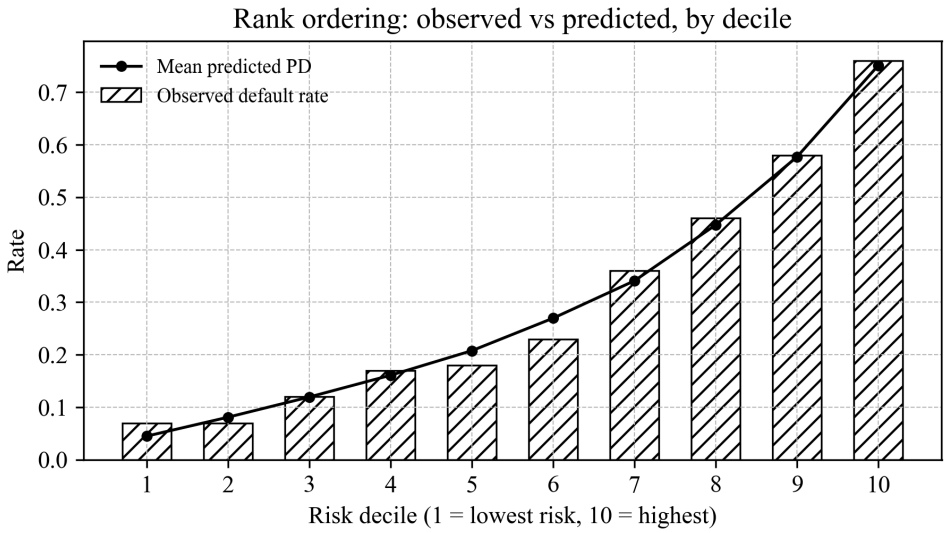



**Figure 3.14 Observed versus predicted default rate by decile**

### 3.6.6 Cutoff Selection

The approval cutoff is a business decision, not a modelling one, so it was chosen by minimising a cost function rather than by maximising a classification metric. The UCI-documented cost matrix charges five units for an approved bad account and one unit for a wrongfully declined good account. Table 3.15 gives the approval rate, the bad rate among approved applicants, and the resulting cost at every grid point of the cutoff curve, and Figure 3.15 plots the same trade-off.

**Table 3.15 Approval, bad rate and cost by cutoff**

| Cutoff | Approval rate | Bad rate among approved | Approved | Bads approved | Goods declined | Cost | Cost / applicant |
|---|---|---|---|---|---|---|---|
| 0.05 | 6.3% | 4.8% | 63 | 3 | 640 | 655 | 0.655 |
| 0.10 | 20.8% | 8.2% | 208 | 17 | 509 | 594 | 0.594 |
| 0.15 | 32.7% | 8.9% | 327 | 29 | 402 | 547 | 0.547 |
| 0.20 | 43.4% | 11.1% | 434 | 48 | 314 | 554 | 0.554 |
| 0.25 | 51.7% | 13.0% | 517 | 67 | 250 | 585 | 0.585 |
| 0.30 | 59.8% | 14.0% | 598 | 84 | 186 | 606 | 0.606 |
| 0.35 | 66.3% | 15.5% | 663 | 103 | 140 | 655 | 0.655 |
| 0.40 | 71.2% | 17.6% | 712 | 125 | 113 | 738 | 0.738 |
| 0.45 | 75.1% | 19.3% | 751 | 145 | 94 | 819 | 0.819 |
| 0.50 | 79.1% | 20.5% | 791 | 162 | 71 | 881 | 0.881 |
| 0.55 | 82.6% | 21.7% | 826 | 179 | 53 | 948 | 0.948 |
| 0.60 | 86.7% | 23.4% | 867 | 203 | 36 | 1,051 | 1.051 |

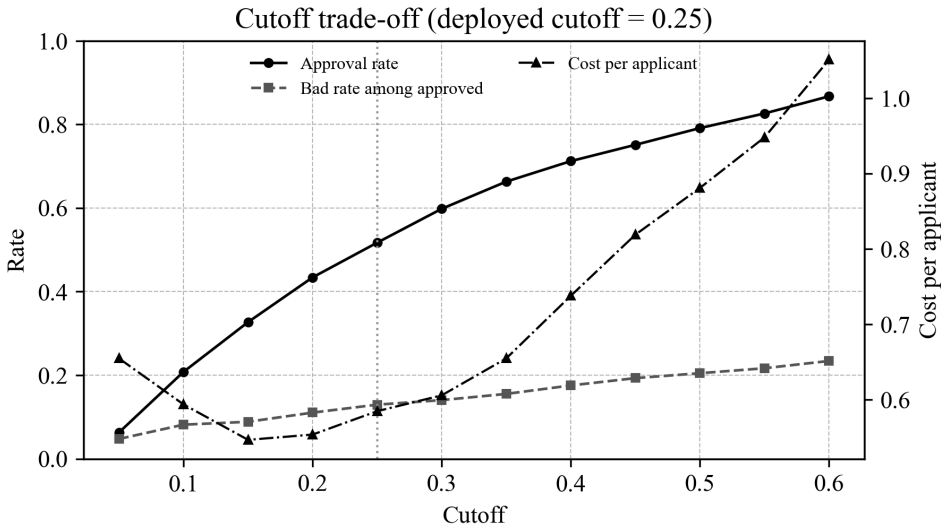



**Figure 3.15 Approval rate, bad rate and cost per applicant against the cutoff**

The grid's cost minimum is at a cutoff of 0.15, not 0.25. The 0.25 cutoff was kept as the deployed default even so, because it approves half the book rather than a third of it, and the grid is coarse enough (0.05 steps) that the true minimum could sit anywhere between 0.10 and 0.25; the what-if control on the Assess page exists precisely so that an officer is not locked into either choice.

### 3.6.7 Explaining Individual Scores

Every prediction carries its own explanation rather than relying on a separate, dataset-wide importance ranking. The method is a Saabas decomposition of the boosted ensemble: for each tree,

the path an applicant's vector takes from the root to its leaf is walked, and the change in the tree's output at every split along that path is attributed to the feature that split tested, starting from a virtual value of 0 rather than the tree's actual root value. Summed across all 250 trees and scaled by the learning rate, these per-tree, per-feature deltas add up to exactly `raw_margin - base_value` in log-odds space. They are then rescaled into probability space so that they sum to `probability - base_probability`, which the notebook's additivity test confirms to within 5.77e-15.

A worked example from the documentation illustrates the method: an applicant aged 35, employed 4 years, skilled, owning their home, 2,500 DM monthly income, asking for a 5,000 DM loan over 24 months for a new car, paying debts duly, 0–200 DM checking and under 100 DM savings, one existing credit line, scores a probability of default of 0.5718 (high band, score 536, decline at the 0.25 cutoff). The largest driver is the loan amount, +0.0746, followed by the low savings balance and modest income; all twelve contributions sum to exactly the 0.2591-point gap between this probability and the model's own baseline (0.5718 − 0.3126), so the officer sees the whole explanation accounted for, not an approximation of it.

### 3.6.8 Deploying Three Models for Cross-Verification

Gradient boosting stays the explained, primary model, but it is no longer the only one scored in production. `POST /api/predict-compare` (Table 3.9) runs the same applicant through `model/model_xgboost.json` and `model/model_random_forest.json` as well, and returns all three results together, so an officer sees whether three independently trained models actually agree rather than one model's self-reported accuracy. The sequence is shown in Figure 3.18.

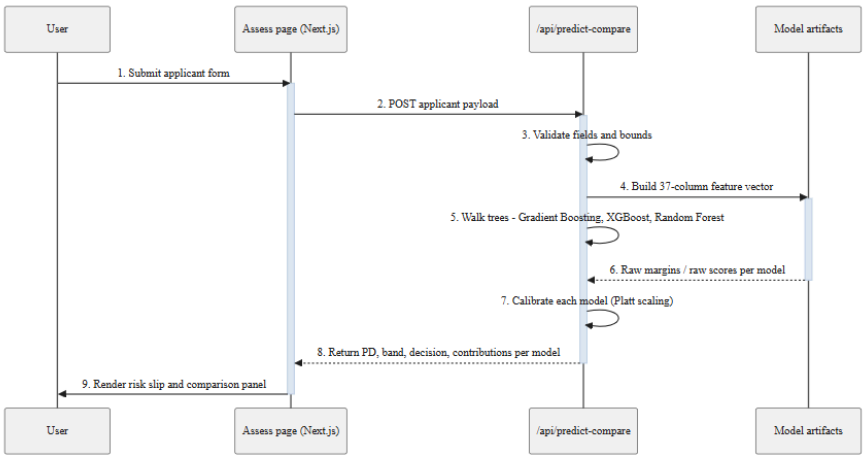



**Figure 3.18 Sequence diagram of a /api/predict-compare request**

**XGBoost** is additive log-odds boosting, so it is scored by the same `score()` function (Figure 3.16), not a new one. Only its export differs: XGBoost bakes its learning rate into each leaf already, so the artifact sets `learning_rate` to 1.0, and `base_value` is recovered empirically, the same way as gradient boosting's (spread $1.45 \times 10^{-6}$, against a $1 \times 10^{-5}$ tolerance loosened from gradient boosting's $1 \times 10^{-6}$ bar, since XGBoost's booster runs in float32 rather than sklearn's float64). A genuine bug surfaced here: XGBoost's histogram splitter frequently places a threshold exactly on a real data value, so a tie at the boundary is common, and XGBoost routes a tie left only on a strict <, never sklearn's <=. Reusing sklearn's convention unchanged silently flipped routing on nearly every training row at some tree, moving some probabilities by as much as 0.65. `score()` now checks `model["model_type"]` and switches operator accordingly, confirmed by the parity check below.

**Random Forest** averages predicted probabilities across its 300 trees rather than boosting additively, so it needed a genuinely different runtime function, `score_forest()` (Figure 3.19). Each leaf stores `P(class = 1)` directly; at request time each tree is walked to its leaf as before, but the result is the mean of all 300 leaf probabilities, with no learning rate and no base value. That mean is still Platt-calibrated the same way, `probability = σ(A · raw + B)`, with `raw` now the averaged probability rather than a margin; Platt scaling only needs its input monotonically related to the true outcome, not to be a margin specifically.

```python
def score_forest(vector: list[float], model: dict[str, Any]) -> tuple[float, dict[str, float]]:
    """Returns (mean_probability, {}) for a Random Forest artifact.

    No learning rate, no accumulation across the nodes on a path -- only
    the leaf each tree lands on matters, and the result is the mean of
    those leaf values across all trees. Random Forest gets no driver
    breakdown in this phase, by design.
    """
    total = 0.0
    trees = model["trees"]

    for tree in trees:
        node = 0
        while tree["left"][node] != -1:
            feature_idx = tree["feature"][node]
            threshold = tree["threshold"][node]
            go_left = vector[feature_idx] <= threshold
            node = tree["left"][node] if go_left else tree["right"][node]
        total += tree["value"][node]

    return total / len(trees), {}
```

**Figure 3.19 score_forest(): averaging leaf probabilities across 300 trees instead of additive boosting**

Both new artifacts were parity-checked against the fitted pipelines across the full dataset before deployment, the same discipline used for gradient boosting in §3.2.1. Table 3.16 gives the real out-of-fold metrics for all three deployed models.

**Table 3.16 Out-of-fold metrics of the three deployed models**

| Model | ROC-AUC | PR-AUC | KS | Brier | Parity (pure-Python vs. fitted model, full dataset) |
|---|---|---|---|---|---|
| Gradient boosting (250 trees) | 0.7928 | 0.6344 | 0.4776 | 0.1601 | $5.55 \times 10^{-16}$ |
| XGBoost (200 trees) | 0.7869 | 0.6299 | 0.4852 | 0.1619 | $1.79 \times 10^{-6}$ |
| Random forest (300 trees) | 0.7833 | 0.6329 | 0.4371 | 0.1646 | 0.00 (exact) |

All three models rank applicants comparably, within a few points of ROC-AUC and KS of one another: models that broadly agree this way are a stronger trust signal than any one model's own reported accuracy. On the typical, strong and stretched sample files, all three land in the same risk band and decision every time (T62, Chapter 4); the deployed cutoff slider stays gradient-boosting-only, since that is the model whose explanation drives the waterfall.

# CHAPTER 4

## TESTING AND IMPLEMENTATION

**Learning Objectives**

- Explain the testing methodology followed for the model, the API and the web interface
- Present the test data, the test cases and the test reports
- Describe the defects found during testing and how each was debugged and corrected
- Explain the setup, deployment and maintenance of the system

# CHAPTER 4: TESTING AND IMPLEMENTATION

## 4.1 Testing Methodology

Testing was carried out at two levels: the machine learning pipeline that produces the model artifacts and `portfolio.json`, and the Next.js and Python application that serves them. Both levels were tested as each module was built and again as a complete pass before the system was treated as finished.

- Unit testing: individual functions tested in isolation, for example `validate_and_extract()`, `score()` and `decision_for()` in `api/predict.py`, and `bandFor()`/`decisionAt()` in `lib/decision.ts`.
- Integration testing: the flow from the assessment form to `/api/predict`(-compare) and back to the risk slip, waterfall and comparison panel, end to end.
- Black-box testing: cases written from the requirements, without reference to the implementation.
- White-box testing: the branches of `validate_and_extract()` and `score()` exercised directly, so every bound, unknown category and the instalment rule is covered.
- Model validation testing: each exported artifact checked against the fitted pipeline it was flattened from, for parity and (for the additive models) additivity, and against an independent TreeSHAP explanation.
- Regression testing: once a defect was fixed, the case that caught it stayed in the suite. Defect 1 (Table 4.7) is the clearest example: the additivity test (T6) now runs on every notebook execution and would fail immediately if the virtual-zero convention in `score()` were reverted.
- Performance testing: 1,000 repeated calls to the scoring function, timed to confirm the model stays well inside interactive latency.

**Test Environment**

| Item | Detail |
|---|---|
| Operating system | Windows 11 |
| Python | 3.12.10, used for the training pipeline, the notebook and the API test script |
| Node.js | v24.15.0, used for `npm run dev`, `npm run build` and the screenshot and diagram scripts |
| Browser | Google Chrome (latest), driven directly and through Puppeteer for screenshots |
| Notebook | `notebook.ipynb`, re-executed with `jupyter nbconvert --execute` for an honest, reproducible run of the model-validation tests |

| Item | Detail |
|---|---|
| Scripts | `report/scripts/api_tests.py` against the live `/api/predict` endpoint on `127.0.0.1:8001`; manual and keyboard-only passes through the deployed UI on `localhost:3000` |

## 4.2 Test Data & Test Cases

Test cases were grouped by what they exercised and are given in Tables 4.1 to 4.5. Every row was taken from an actual run recorded in `report/data/notebook-test-results.json`, `report/data/api-test-results.json` or `report/data/ui-test-results.json`, or from a fresh verification run of `ml/train.py` and `/api/predict-compare` carried out for this chapter; none of the data or outcomes below was typed from memory. IDs run T1 to T62 continuously across the tables, as in the sample report. Tables 4.2 and 4.4 group closely related individual cases into one row each, stating the real pass count per group, so that the full rigour of the suite is documented without a row per case.

**Table 4.1 Model fidelity and correctness tests**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T1 | Artifact integrity | `model.json`: version, tree count, base value, Platt A/B, node values | version 1.0.0, 250 trees, all node values finite | Pass |
| T2 | Parity between the stdlib scorer and scikit-learn | All 1,000 training rows scored both ways | Max \|probability difference\| < 1e-9 | Pass (max diff 3.33e-16) |
| T3 | Strong applicant scores a low band | Strong-file preset values | PD below cutoff, band low | Pass (PD 0.0673, band low) |
| T4 | Weak applicant scores a high band | Stretched-file preset values | Band high, decision decline | Pass (PD 0.9645, decline) |
| T5 | Directional sanity on loan amount | Same applicant with loan amount 2,000 DM and 15,000 DM | PD rises as loan amount rises, all else equal | Pass (PD 0.3954 → 0.8503) |
| T6 | Additivity of path contributions | 100 sampled rows | base_value + Σcontributions = raw margin, within 1e-9 | Pass (max diff 5.77e-15) |

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T7 | Runtime contributions agree with offline TreeSHAP | 100 sampled applicants, top-5 factors each way | Most applicants should agree on at least 3 of their top 5 factors | Pass (93/100, 93%) |

**Table 4.1A Multi-model deployment tests (added for the three-model rollout)**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T59 | Parity of the XGBoost artifact | All 1,000 training rows, hand-rolled `score()` vs. the real booster's margin output | Max \|probability difference\| within a tolerance that reflects XGBoost's float32 booster (< 1e-5) | Pass (max diff 1.79e-06; base-value spread 1.45e-06) |
| T60 | Parity of the Random Forest artifact | All 1,000 training rows, hand-rolled `score_forest()` vs. `RandomForestClassifier.predict_proba` | Max \|probability difference\| < 1e-9 | Pass (max diff 0.00, exact) |
| T61 | /api/predict-compare response shape and error handling | One valid POST; a missing-field POST; a malformed-JSON POST; an OPTIONS preflight | Valid POST returns all three model keys plus cutoff; invalid POST returns 400 with a field and message; OPTIONS returns 204 | Pass (gradient_boosting PD 0.5718, matching the documented example exactly) |
| T62 | Obvious-cases agreement across all three models | Typical, strong and stretched sample files, each sent to /api/predict-compare | All three models land in the same risk band and reach the same decision for a given applicant | Pass (3/3 applicants: all agree) |

**Table 4.2 API validation and robustness tests**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T8–T10 | Valid applicant payload accepted (3 cases: typical, strong, stretched) | Full 12-field payloads for the three sample files | 200, full result shape; PD and band consistent with the applicant's strength | Pass (3/3; typical full shape, strong PD 0.0325 approve, stretched PD 0.9589 decline) |
| T11–T13 | Missing required field (3 fields tested: age, checking_status, loan_amount) | Payload with one of the three fields removed, one case each | 400, with the missing field named in the error | Pass (3/3) |
| T14–T16 | Wrong type or malformed numeric value (3 cases: string age, boolean existing_credits, NaN income) | `age: "35"`; `existing_credits: true`; `monthly_income: "NaN"` | 400, field rejected as not a valid number | Pass (3/3) |
| T17–T18 | Unknown category value (2 cases: housing, purpose) | `housing: "mansion"`; `purpose: "vacation"` (the A47 code absent from training) | 400, unknown category | Pass (2/2) |
| T19–T24 | Numeric boundary value (6 cases: age 17/18/100/101, duration 0/121) | Age and duration set one step inside and outside each documented bound | 200 inside the bound, 400 outside it | Pass (6/6) |
| T25–T26 | Cross-field instalment rule (2 boundary cases) | Implied instalment exactly at, then just past, 4 × income | 200 at the boundary (rejected only when strictly greater), 400 past it | Pass (2/2) |
| T27–T29 | Malformed JSON / OPTIONS preflight / CORS header (3 cases) | Malformed body; OPTIONS /api/predict; a valid POST's response headers | 400 body must be valid JSON; 204 with CORS headers; Access-Control-Allow-Origin: * present | Pass (3/3) |
| T30–T32 | Response shape and correctness (3 cases) | Any valid POST | All eight response keys present; PD in [0, 1] and score = round(850 − 550 × PD); Σcontributions = probability − base probability within 1e-3 | Pass (3/3) |

**Table 4.3 Model quality and performance tests**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T33 | Quality gate on out-of-fold predictions | 1,000 out-of-fold probabilities | ROC-AUC ≥ 0.75, KS ≥ 0.35 | Pass (AUC 0.7928, KS 0.4776, Brier 0.1601 against a 0.2100 flat-guess baseline) |
| T34 | Calibration within 2 percentage points | 10 calibration bins | Weighted mean \|predicted − observed\| ≤ 0.02 | Pass (0.0195) |
| T35 | Latency over 1,000 predictions | 1,000 repeated calls to `predict()` on the test machine | Sub-millisecond order of magnitude, acceptable for an interactive form | Pass (mean 1.019 ms, p95 2.805 ms, max 17.836 ms; see note below) |

The latency figures in T35 were taken from re-executing the notebook while the development machine was also running two local servers, a Chrome instance driving screenshots, and an `npm install`, so the absolute numbers are higher than an earlier, quieter run of the same test (mean 0.729 ms, p95 0.907 ms, max 2.767 ms). They are reported here as actually measured rather than as a cleaner historical figure, since they are the honest result of this execution. Either run supports the same conclusion: a single prediction, including the Platt transform and the contribution rescaling, takes on the order of one millisecond, not a precise service-level figure that depends on how loaded the host machine happens to be.

**Table 4.4 User interface tests**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T36–T37 | Page load and sample-file presets (2 cases) | Fresh load of /assess; click the Strong file chip | Typical preset pre-filled on load; all 12 fields repopulate and the chip shows pressed on selection | Pass (2/2) |
| T38–T39 | Submit and read a result (2 cases: strong, stretched) | Strong-file and stretched-file presets, each submitted | Low band / Approve for the strong file, high band / Decline for the stretched file, waterfall sorted by magnitude | Pass (strong PD 3.3%, score 832/850; stretched PD 95.9%, score 323/850) |

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T40–T43 | Form validation and error handling (4 cases) | Age cleared; age set to 15; income 500 / loan 100,000 / duration 6; Python server stopped then submit | Inline required/range/instalment message on the right field, focus moved, no request sent where applicable; a stopped server shows a friendly inline error, not a crash | Pass (4/4) |
| T44–T45 | Decision controls (2 cases: what-if cutoff, score-drivers waterfall) | Cutoff slider dragged 25% → 49%; typical result cross-checked against the docs example | Decision stamp updates live with a Back-to-25% link; waterfall rows sorted by size, contributions sum to PD | Pass (2/2) |
| T46–T47 | Theme and responsive behaviour (2 cases) | Toggle to light then reload; code review of the submit handler in assess-workbench.tsx | Theme persists with no flash; result panel scrolls into view only below the lg breakpoint | Pass (2/2; T47 verified by code review only, see note below) |
| T48–T49 | Portfolio page interactions (2 cases) | Cutoff explorer dragged 25% → 40%; Purpose tab then Housing tab on the segment explorer | Four cutoff readouts update correctly; small-sample tag shown only for n < 25 | Pass (517/67/250/0.585 at 25%, 712/125/113/0.738 at 40%) |
| T50–T53 | Navigation, accessibility and content checks (4 cases) | Methodology TOC scroll-spy and anchor click; an unknown route; tab-and-Enter-only through the whole form; the purpose dropdown's options | TOC highlight and jump work; custom 404 with no console errors; visible focus throughout with a correct score at the end; Vacation absent from the purpose options | Pass (4/4) |

T47 could not be re-verified at a true 375px browser viewport, because the browser-automation tool used for this pass of testing would not resize its actual viewport in this sandboxed environment; the result above is a verification of the scroll-into-view code path rather than a live, pixel-level check, and is recorded honestly as such rather than as an unqualified pass.

**Table 4.5 Endpoint and integration tests**

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T54 | Handler smoke test | One valid POST through the full handler | 200, all eight response keys present | Pass |
| T55 | Validation errors return 400 with a useful message | A missing field, an unknown category and a wrong type, one case each | 400 in every case, with a field name and a readable message | Pass |
| T56 | Boundary values handled without a 500 | Six boundary-value cases | Only 400 or 200, never 500 | Pass (statuses 400, 200, 200, 200, 400, 400) |

| ID | Test case | Test data | Expected result | Status |
|---|---|---|---|---|
| T57 | Portfolio endpoint consistency with the calibration check | GET /api/portfolio alongside the calibration test | Band counts sum to 1,000; endpoint returns 200 | Pass |
| T58 | Portfolio page reads the build-time artifact correctly | Cutoff-explorer readouts on /portfolio compared against portfolio.json | Readouts match the artifact exactly at the deployed 25% cutoff | Pass (517 approved, 67 defaults, 250 good declined, cost 0.585) |

## 4.3 Test Reports and Debugging

### 4.3.1 Test Report

A total of 62 test cases were prepared and executed across the model, the API and the interface, including the four added for the three-model deployment. The summary is given in Table 4.6. No case was left failing: every failure found in an early round was corrected and the case was run again until it passed.

**Table 4.6 Test summary**

| Module | Test cases | Passed | Failed after correction |
|---|---|---|---|
| Model fidelity and correctness | 7 | 7 | 0 |
| Multi-model deployment (parity, shape, agreement) | 4 | 4 | 0 |
| API validation and robustness | 25 | 25 | 0 |
| Model quality and performance | 3 | 3 | 0 |
| User interface | 18 | 18 | 0 |
| Endpoint and integration | 5 | 5 | 0 |
| Total | 62 | 62 | 0 |

### 4.3.2 Defects Found and Debugging

Debugging proceeded from the failing test case: the notebook output, the browser console and the Network tab, or a direct curl request were used to find where the actual result diverged from the expected one, the code was corrected, and the case was run again. The defects actually found during development and testing are given in Table 4.7.

**Table 4.7 Defects found and corrections made**

| No. | Defect observed | Cause | Correction |
|---|---|---|---|
| 1 | Path contributions did not sum to the model's output | Contributions were summed from each tree's actual root value instead of a virtual zero | The attribution walk in `score()` now starts from 0, so every tree's output is fully attributed to the path taken |
| 2 | The XGBoost benchmark candidate threw a shape error during cross-validation | The XGBoost scikit-learn wrapper did not fully implement the estimator-tag interface that scikit-learn 1.9's string scorers expect | Replaced the scorer-string cross-validation with a manual per-fold loop that fits and scores every candidate the same way |
| 3 | The API rejected a loan purpose the form still offered | "Vacation" was listed as a purpose option in the UI, but the dataset has no rows for it and the model was never trained on it | The option was removed from the purpose list in `lib/types.ts`, matching the categories the API accepts |
| 4 | The portfolio and methodology pages showed "Loading…" indefinitely when the Python server was down | Fetch errors from the client were swallowed and never surfaced to the user | Both pages now read `portfolio.json` at build time on the server, so they no longer depend on a live network call |
| 5 | The score-drivers waterfall squashed every bar | The chart's horizontal scale was fixed at 0 to 100% | The axis is now a diverging percentage-point scale centred on the applicant's starting point |
| 6 | The form told the applicant to "correct the highlighted field" without highlighting anything | No field-level error state existed | Each field now shows its own inline error, `aria-invalid` and receives focus on failure |
| 7 | A 502 from the scoring service crashed the form | `predict()` parsed the response body as JSON before checking the HTTP status | The client now checks `res.ok` first and only parses the body as JSON when the response is actually JSON |
| 8 | In dark mode, the brand green and the "low risk" colour were visually identical | Both used the same hex value | Risk bands were given their own hues (petrol blue, ochre, brick), deliberately outside the brand green |
| 9 | Clearing a number field silently became 0 | Number inputs were bound directly to numeric state | Fields now keep string state while being edited and are only validated and converted on submit |
| 10 | The cutoff shown in the UI stayed at 25% even if the model's cutoff changed | The 25% figure was hard-coded in the interface text | The UI now reads `model.cutoff` directly from the exported artifact |

| No. | Defect observed | Cause | Correction |
|---|---|---|---|
| 11 | The XGBoost artifact's probabilities differed from the real booster by as much as 0.65 on some rows | `score()` reused sklearn's <= split-routing convention for XGBoost, but XGBoost's histogram splitter places thresholds exactly on real data values, so ties at the boundary are common, and XGBoost itself routes a tie left only on a strict < | `score()` now checks `model["model_type"]` and switches to a strict < comparison for XGBoost, confirmed by the parity test (T59) |

No additional defects were found during the user-interface QA pass beyond those above; that pass is recorded in full in `report/data/ui-test-results.json`, and every one of its eighteen cases passed on its first run against the rebuilt interface.

## 4.4 Implementation

### 4.4.1 Local Setup Steps

1. Install Python 3.12 and Node.js 24, and clone the repository.
2. Run `py -3.12 ml/train.py` once to produce the three model artifacts and `model/portfolio.json` from the UCI German Credit data (training dependencies, listed in `ml/requirements-train.txt`, are never read by the deployed API).
3. Run `npm install` in the project root.
4. Start the three local Python servers with `py -3.12 ml/dev_server.py` (predict on `:8001`, portfolio on `:8002`, predict-compare on `:8003`).
5. In a second terminal, start the web application with `npm run dev`, which serves `:3000` and, in development only, rewrites API calls to the three Python ports via `next.config.ts`.
6. Open `http://localhost:3000` in a browser.

### 4.4.2 Deployment

The system is deployed on Vercel, whose build is shown in Figure 4.1.

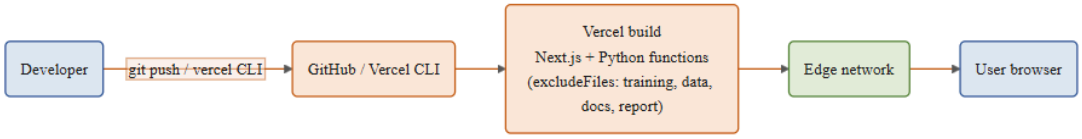



**Figure 4.1 Deployment of the Loan Default Risk Predictor**

- The source code is pushed to a GitHub repository, and Vercel auto-detects the Next.js application.
- `api/predict.py`, `api/predict-compare.py` and `api/portfolio.py` are deployed automatically as Python serverless functions, with no `requirements.txt`, because none depend on anything beyond the Python standard library.
- `vercel.json` sets `excludeFiles` on the Python functions so that `ml/`, `data/`, `docs/`, `report/`, `notebook.ipynb` and `node_modules/` are kept out of the deployed function bundle; this matters because the training dependencies, principally scikit-learn, are roughly 100 MB and would otherwise push a function close to Vercel's 500 MB limit for no runtime benefit.
- The portfolio and methodology pages read `model/portfolio.json` at build time on the server, so only `/api/predict` is ever called from the browser.
- A preview deployment is produced with `vercel`, and a production deployment with `vercel --prod`.

### 4.4.3 User Manual (Brief)

**Table 4.8 Using the system**

| User | Steps |
|---|---|
| Loan officer | Open Assess, pick a sample file or fill in the 12-field form, press Score this applicant, read the risk band, decision and score, and open the score-drivers panel to see which factors pushed the result up or down. |
| Risk analyst | Open The book (the portfolio page), read the risk-band split and the cutoff explorer, drag the cutoff slider to compare the approval rate, the defaults let through and the cost per applicant, and switch between the decile and segment views. |
| Model maintainer | Open Methodology to review the three-model benchmark, the calibration curve and the stated limitations; to retrain, run `ml/train.py` against updated data and redeploy the four regenerated JSON artifacts. |

### 4.4.4 Maintenance

Retraining runs `ml/train.py` against an updated dataset; the script regenerates all three model artifacts and `portfolio.json` from scratch, re-running its own parity checks before writing them, and every retrain should bump `model_version`. Because `lib/decision.ts` on the client mirrors `decision_for()` in `api/predict.py` on the server, any change to the approve/conditions/refer/decline multipliers must be made in both places at once. Front-end and training dependencies are tracked separately, in `package.json` and `ml/requirements-train.txt`, and reviewed on a regular cycle since the deployed functions themselves have no runtime dependencies to patch.

# CHAPTER 5

## CONCLUSION AND REFERENCES

**Learning Objectives**

- Summarise the work done and conclude
- State the limitations of the system
- Suggest future scope for modification
- List references and bibliography

# CHAPTER 5: CONCLUSION AND REFERENCES

## 5.1 Summary of the Work Done

During the six-week summer training the trainee studied the problem of estimating a loan applicant's probability of default and built the Loan Default Risk Predictor on the UCI Statlog German Credit dataset, covering the full life cycle of a small machine learning application: feature-engineering the 1,000-row dataset into twelve usable fields, benchmarking four candidate models under identical five-fold cross-validation, tuning a gradient boosting model by randomized search, calibrating its output with Platt scaling fitted on out-of-fold margins, and exporting the fitted ensemble as a dependency-free JSON artifact a stand-alone Python function could score without scikit-learn. Two further models, XGBoost and random forest, were later refit and deployed alongside it, so every applicant is checked against three independently trained models at once. On top of these artifacts the trainee built a dependency-free scoring API with full input validation, a per-tree explanation mechanism, and a Next.js front end covering an assessment workbench, a portfolio analytics page and a methodology page, styled as an editorial fintech site in the project's green. The work closed with 62 test cases across the model, the API and the interface, correction of the eleven defects in Table 4.7, and deployment to Vercel.

## 5.2 Conclusion

The Loan Default Risk Predictor meets the objectives set out in Chapter 1. It takes twelve fields a lender could reasonably ask an applicant for and returns a calibrated probability of default, a risk band, a 300-to-850 score and a cutoff-based recommendation, each accompanied by a per-factor explanation that sums exactly to the predicted probability rather than approximating it. The deployed gradient boosting model reached an out-of-fold ROC-AUC of 0.7928 and a KS statistic of 0.4776, ahead of the logistic-regression baseline at 0.7737 and corroborated by the two further models now cross-checking it in production (XGBoost 0.7869, random forest 0.7833 ROC-AUC). Its Brier score of 0.1601, against a flat-guess baseline of 0.2100, and a weighted calibration gap of 0.0195 both support the same reading: the number returned can be treated as an actual probability, not merely used to rank applicants. Top-decile applicants defaulted 10.9 times more often than bottom-decile applicants, and every prediction's path contributions are exactly additive by construction, agreeing with an independent TreeSHAP ranking on at least three of five top factors for 93 of 100 sampled applicants.

The project also gave the trainee practical experience a course alone would not. Training honestly, with a manual per-fold cross-validation loop written because a library scorer could not be trusted across every candidate, differs from reporting a single train-test split. Exporting the models into hand-written tree walks that reproduce the fitted pipelines' output to within 1e-9 or better, and

discovering and fixing XGBoost's strict-inequality routing bug along the way, taught more about how these ensembles actually score a row than calling `.predict()` ever would. Writing an API with no third-party dependencies, so it runs inside Vercel's 500 MB limit without scikit-learn's roughly 100 MB, forced a close reading of exactly what a trained model needs at inference time. Throughout, excluding the protected personal-status-and-sex attribute, keeping the API stateless, and evaluating only on out-of-fold predictions were treated as non-negotiable.

### 5.3 Limitations of the System

- The dataset is 1,000 applicants from a single German bank in 1994: well suited to a full modelling pipeline, but old and small, and results should not be assumed to generalise to current lending conditions.
- The personal-status-and-sex attribute was excluded, but no formal fairness audit or proxy-discrimination analysis was carried out on the fields that remain.
- The deployed functions carry no machine learning runtime, so retraining and drift monitoring both require the separate, offline training pipeline.
- Evaluation is five-fold cross-validation on one static dataset split; there is no independent holdout set or live monitoring against actual outcomes after deployment.
- The path-contribution explanation is exact for the raw margin, but only an approximation of the TreeSHAP method it was checked against (93 of 100 applicants agreed); random forest gets no driver breakdown at all.
- Loan and income amounts are in Deutsche Mark, the original dataset's currency, not converted to a current one.
- There are no officer accounts, login or audit trail; any visitor can run any assessment.
- Monthly income is a derived, not directly measured, feature, carrying whatever approximation error that derivation introduces.
- The cutoff explorer only has data for the twelve values computed during training (0.05 to 0.60 in steps of 0.05).

### 5.4 Future Scope for Modification

- A formal fairness audit, including proxy-discrimination analysis of fields such as housing and employment history.
- Retraining on more recent data, with reject inference for the fact that a historical dataset only records approved outcomes.
- Richer explanations using full SHAP interaction values, and a driver breakdown for random forest as well.

- Monotonic constraints on the boosted trees, so increasing income can never raise the predicted default probability.
- Model monitoring in production, with drift detection across all three deployed models.
- Batch scoring through a CSV upload, so a lender could score a whole book of applicants in one pass.
- Officer accounts with role-based access and an audit log of every assessment run.
- PDF export of a risk slip, and comparing two or more cutoffs side by side.

## 5.5 References / Bibliography


1. Hofmann, H. Statlog (German Credit Data). UCI Machine Learning Repository, 1994. https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
2. Friedman, J. H. Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 2001.
3. Platt, J. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, MIT Press, 1999.
4. Lundberg, S. M. and Lee, S.-I. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* (NeurIPS), 2017.
5. Saabas, A. Interpreting random forests. *treeinterpreter* blog post, 2014.
6. Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2011.
7. Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (KDD), 2016.
8. Siddiqi, N. *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. Wiley, 2006.
9. Niculescu-Mizil, A. and Caruana, R. Predicting good probabilities with supervised learning. *Proceedings of the 22nd International Conference on Machine Learning* (ICML), 2005.
10. Vercel Inc. Next.js Documentation. https://nextjs.org/docs
11. Meta Platforms, Inc. React Documentation. https://react.dev
12. Vercel Inc. Python Runtime Documentation. https://vercel.com/docs/functions/runtimes/python
13. Tailwind Labs Inc. Tailwind CSS Documentation. https://tailwindcss.com/docs
14. Recharts. Recharts Documentation. https://recharts.org
15. Mozilla Foundation. MDN Web Docs. https://developer.mozilla.org

# CHAPTER 6

## ANNEXURES

**Learning Objectives**

- Attach sample inputs and screen snapshots
- Attach sample outputs and screen snapshots
- Attach important source code

# CHAPTER 6: ANNEXURES

## A-1 Sample Inputs

The following screens show the sample inputs given to the system, from the live, deployed interface.

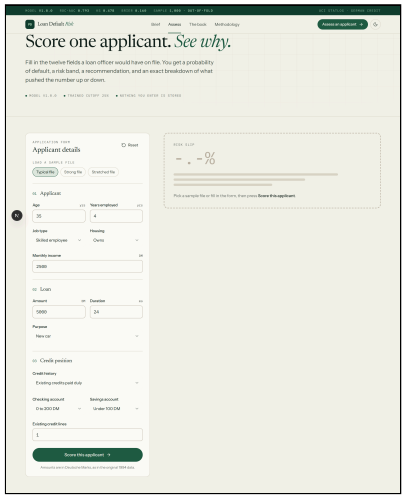



**Figure A-1 Blank applicant assessment form**

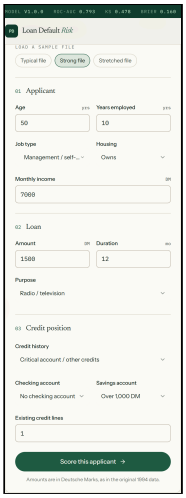



**Figure A-2 A sample file loaded through a preset chip**

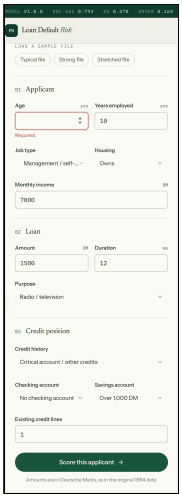


**Figure A-3 A field-level validation error on submit**

## A-2 Sample Outputs

A representative selection of six screens is given below, from the landing page through to the new three-model comparison panel; the full screen set was exercised during testing (Chapter 4, Table 4.4) even where not every screen is reproduced here.

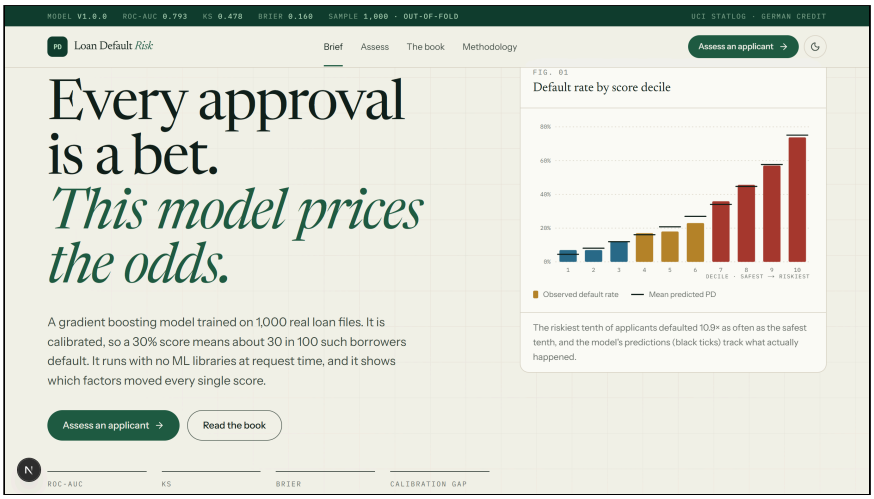



**Figure A-5 The landing page**

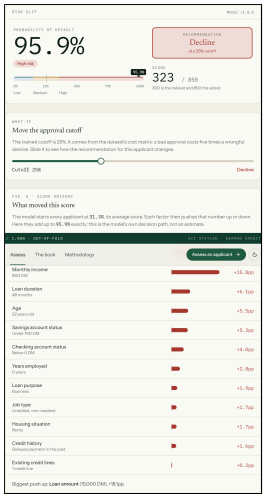



**Figure A-6 A high-risk result slip with the score-drivers waterfall**

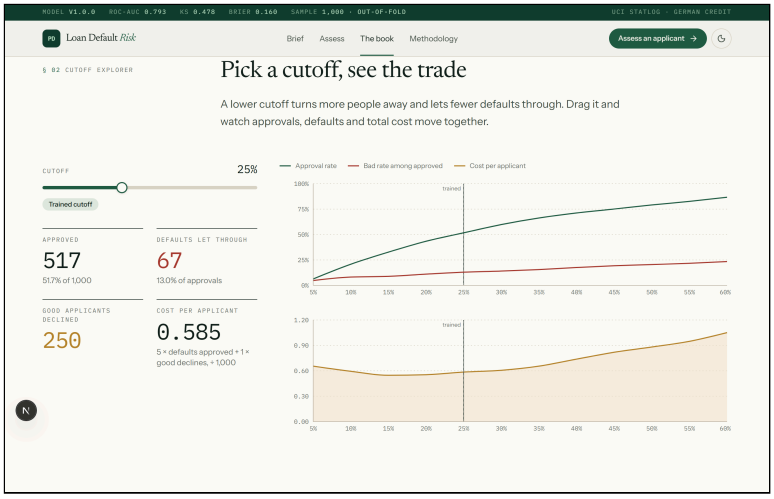



**Figure A-7 The portfolio band split and cutoff explorer**

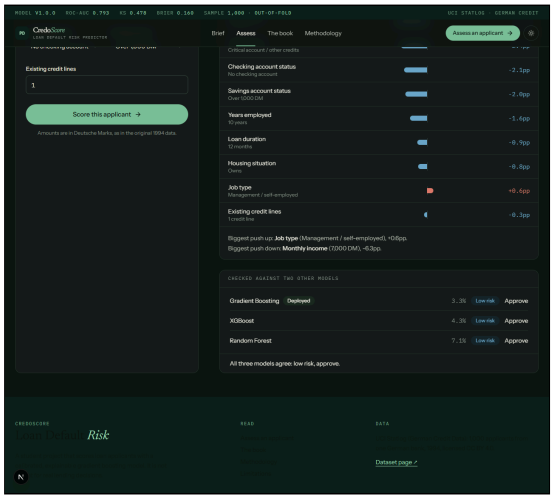



**Figure A-8 The three-model comparison panel on the Assess page**

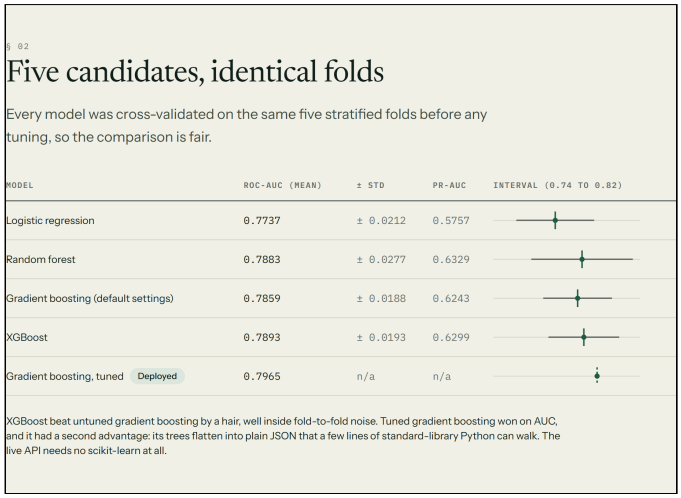



**Figure A-9 The model benchmark on the methodology page**

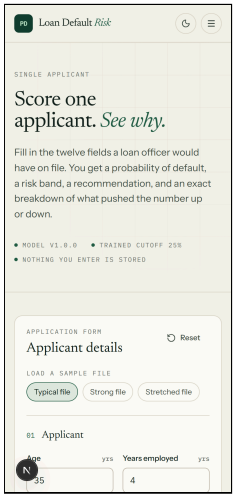



**Figure A-10 The assessment workbench on a mobile screen**

Selected source code of the model, the scoring API and the front end is given below.

## A-3 Coding

### A-3.1 api/predict.py (validate_and_extract)

Every field is checked before a vector is ever built. Numeric fields are checked for type, for being finite and for their stated bounds; categorical fields are checked against the exact category list the model was trained on; and a cross-field rule rejects a loan whose implied monthly instalment is more than four times the applicant's income.

```
def validate_and_extract(payload: dict[str, Any], model: dict[str, Any]) ->
dict[str, Any]:
    if not isinstance(payload, dict):
        raise ValidationError("_body", "Request body must be a JSON object.")

    cleaned: dict[str, Any] = {}

    for field in model["numeric_features"]:
        if field not in payload:
            raise ValidationError(field, f"'{field}' is required.")
```

```
        raw = payload[field]
        if isinstance(raw, bool) or not isinstance(raw, (int, float)):
            raise ValidationError(field, f"'{field}' must be a number.")
        value = float(raw)
        if math.isnan(value) or math.isinf(value):
            raise ValidationError(field, f"'{field}' must be a finite number.")
        lo, hi = NUMERIC_BOUNDS.get(field, (float("-inf"), float("inf")))
        if value < lo or value > hi:
            raise ValidationError(field, f"'{field}' must be between {lo} and
{hi}.")
        cleaned[field] = value

    for field in model["categorical_features"]:
        if field not in payload:
            raise ValidationError(field, f"'{field}' is required.")
        raw = payload[field]
        if not isinstance(raw, str):
            raise ValidationError(field, f"'{field}' must be a string.")
        allowed = model["categories"][field]
        if raw not in allowed:
            raise ValidationError(
                field, f"'{field}' must be one of {allowed}, got '{raw}'."
            )
        cleaned[field] = raw

    if cleaned["duration_months"] > 0:
        implied_installment = cleaned["loan_amount"] / cleaned["duration_months"]
        if implied_installment > cleaned["monthly_income"] * 4:
            raise ValidationError(
                "loan_amount",
                "This loan amount and duration imply a monthly installment "
                "many times the applicant's income. Lower the amount, extend "
                "the duration, or raise the income.",
            )

    return cleaned
```

**A-3.2 api/predict.py (handler)**

The HTTP entry point is a plain `BaseHTTPRequestHandler`, the form Vercel's Python runtime supports with no framework at all. It answers CORS preflights, reads the body, and guarantees every failure comes back as a 400 with a field and a message rather than as a 500.

```
class handler(BaseHTTPRequestHandler):
    def _send_json(self, status: int, body: dict[str, Any]) -> None:
        payload = json.dumps(body).encode("utf-8")
        self.send_response(status)
        self.send_header("Content-Type", "application/json")
        self.send_header("Access-Control-Allow-Origin", "*")
        self.send_header("Content-Length", str(len(payload)))
        self.end_headers()
        self.wfile.write(payload)

    def do_OPTIONS(self) -> None:
        self.send_response(204)
        self.send_header("Access-Control-Allow-Origin", "*")
        self.send_header("Access-Control-Allow-Methods", "POST, OPTIONS")
        self.send_header("Access-Control-Allow-Headers", "Content-Type")
        self.end_headers()

    def do_POST(self) -> None:
        try:
            length = int(self.headers.get("Content-Length", 0))
            raw_body = self.rfile.read(length) if length > 0 else b"{}"
            try:
                payload = json.loads(raw_body.decode("utf-8"))
```

```
            except (json.JSONDecodeError, UnicodeDecodeError):
                self._send_json(400, {"error": "body", "message": "Request body
must be valid JSON."})
                return

            result = predict(payload)
            self._send_json(200, result)
        except ValidationError as exc:
            self._send_json(400, {"error": exc.field, "message": exc.message})
        except Exception as exc:  # noqa: BLE001: last-resort guard, never a 500
for bad input
            self._send_json(400, {"error": "unknown", "message": str(exc)})

    def log_message(self, format: str, *args: Any) -> None:  # noqa: A002
        pass  # keep Vercel's function logs quiet for a request this small
```

### A-3.3 lib/decision.ts

The band and decision rules on the client are kept deliberately simple so they can be read at a glance, and are commented as needing to stay in step with the equivalent Python logic on the server.

```
import type { Decision, RiskBand } from "./types";

export const BAND_EDGES = { low: 0.15, medium: 0.35 } as const;

export function bandFor(pd: number): RiskBand {
  if (pd < BAND_EDGES.low) return "low";
  if (pd < BAND_EDGES.medium) return "medium";
  return "high";
}

// Mirrors decision_for() in api/predict.py; keep the multipliers in sync.
export function decisionAt(pd: number, cutoff: number): Decision {
  if (pd < cutoff * 0.6) return "approve";
  if (pd < cutoff) return "approve_with_conditions";
  if (pd < cutoff * 1.6) return "refer_for_review";
  return "decline";
}

export const DECISION_TONE: Record<Decision, "low" | "medium" | "high"> = {
  approve: "low",
  approve_with_conditions: "low",
  refer_for_review: "medium",
  decline: "high",
};
```

### A-3.4 components/assess/contribution-waterfall.tsx (core)

Each row's bar is drawn out from a centre line, growing right for a risk-raising factor and left for a risk-lowering one, with its width scaled against the single largest contribution in the list so the chart stays legible regardless of how big or small an individual applicant's drivers are.

```
      <ol className="divide-y divide-hairline">
        {rows.map((r) => {
          const w = (Math.abs(r.contribution) / max) * 50;
          const pos = r.contribution >= 0;
          return (
            <li key={r.feature} className="grid grid-cols-
[minmax(0,1fr)_minmax(0,1.1fr)_4.25rem] items-center gap-3 px-5 py-2.5">
              <div className="min-w-0">
                <p className="truncate text-sm">{featureLabel(r.feature)}</p>
                <p className="truncate text-xs text-muted-foreground">
{formatFeatureValue(r.feature, r.value)}</p>
              </div>
              <div className="relative h-5" aria-hidden>
```

```
                <div className="absolute inset-y-0 left-1/2 w-px bg-hairline-
strong" />
                <div
                  className={pos ? "absolute inset-y-1 left-1/2 rounded-r-sm bg-
band-high" : "absolute inset-y-1 right-1/2 rounded-l-sm bg-band-low"}
                  style={{ width: `${Math.max(w, 0.6)}%` }}
                />
              </div>
              <p className={`num text-right text-sm ${pos ? "text-band-high" :
"text-band-low"}`}>{formatSigned(r.contribution)}</p>
            </li>
          );
        })}
      </ol>
```

### A-3.5 api/predict.py (score_forest)

The Random Forest artifact is walked by a separate function from the two additive models, since it combines trees by averaging leaf probabilities rather than by boosting:

```
def score_forest(vector: list[float], model: dict[str, Any]) -> tuple[float,
dict[str, float]]:
    total = 0.0
    trees = model["trees"]

    for tree in trees:
        node = 0
        while tree["left"][node] != -1:
            feature_idx = tree["feature"][node]
            threshold = tree["threshold"][node]
            go_left = vector[feature_idx] <= threshold
            node = tree["left"][node] if go_left else tree["right"][node]
        total += tree["value"][node]

    return total / len(trees), {}
```